

PMATH 351: Real Analysis with Professor Alexandru Nica

This course is fundamentally an introduction to analysis of metric spaces, which bring an axiomatic approach to the notion of distance between points of a space.

Lecture 0: Countable and Uncountable Sets:

Definition 0.1:

An infinite countable set is a set A which can be put into bijective correspondence with the set $\mathbb{N} = \{1, 2, 3, \dots\}$ of positive integers, equivalently, we can write it in an infinite sequence with no repetitions. A set is called countable if it is infinite countable or finite (including empty).

Proposition 0.2:

Let A be countable. Then $B \subseteq A$ is countable.

Proof:

There are three cases:

If A is finite, then its subset B is also finite and thus countable.

If A is infinite and B is finite, then B is countable. Therefore suppose B is infinite. Because A is infinite countable, write it as the sequence $(a_1, a_2, \dots)$. Because B is a subset that is infinite, it must form an (infinite) subsequence of this sequence $B = \{a_{m(1)}, a_{m(2)}, \dots\}$ for $1 \leq m(1) < m(2) < \dots$, but the function $g : \mathbb{N} \rightarrow B$ defined by $g(k) = a_{m(k)}$ is a bijection, hence B is countable. ■

Corollary 0.3:

Let A be nonempty and countable. Suppose that C is nonempty and there exists an injective function $f : C \rightarrow A$. Then C is countable.

Proof:

Let $B = f(C) = \{a \in A \mid \exists c \in C \text{ s.t. } f(c) = a\} \subseteq A$. By the previous proposition, B is countable. But because f is injective, it gives a bijection from C to B . This implies C is countable, since either B is finite (so the bijection ensures C is finite) or it is infinite countable, in which case there is a string of bijections $C \xrightarrow{\cong} B \xrightarrow{\cong} \mathbb{N}$ which means C is in bijection with $\mathbb{N}$, hence infinite countable. ■

Note: We can make an analogous statement to Corollary 0.3 for surjective functions, which is Problem P0.2 and will be on the homework.

Exercise 0.4:

$\mathbb{N} \times \mathbb{N}$ is countable. This can be done by choosing a sequence of its points in the lattice $\mathbb{N} \times \mathbb{N}$ that picks off the anti-diagonals when writing out $\mathbb{N} \times \mathbb{N}$. Alternatively use the injection $f((m, n)) = 2^m 3^n$ which is injective by the uniqueness of prime factorization, which is countable by the above corollary ■.

Exercise 0.5:

Prove $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Remark 0.6:

Cantor: $\mathbb{R}$ is not countable. We will see this proof next lecture.

Lecture 0 continued: More uncountable and countable sets

Last time we discussed the definition and basic examples of countable sets, and the “injective tool” which lets us prove some sets are countable by injecting them into a countable set. There is also a “surjective tool” to be proved on the next homework, where we surject a countable set into another set, proving the other set is countable.

Now we wish to prove that countable unions “preserve” countability:

Proposition 0.8:

Let $B_1, B_2, \dots, B_n, \dots$ be countable. Then $\bigcup_{m=1}^{\infty} B_m$ is countable as well.

Let us use a lemma to prove this:

Lemma 0.7:

Let $A_1, \dots, A_n, \dots$ be countable and pairwise disjoint ($A_i \cap A_j = \emptyset$ for $i \neq j$ in $\mathbb{N}$). Then $\bigcup_{i=1}^{\infty} A_i$ is countable.

Proof:

Let $A := \bigcup_{i=1}^{\infty} A_i$. Assume without loss of generality that A is nonempty (if $A = \emptyset$, we are done). Let $M = \{m \in \mathbb{N} \mid A_m \neq \emptyset\}$. Our idea is to use the injective tool to inject A into $\mathbb{N} \times \mathbb{N}$. For every $m \in M$, we pick an injective function $f_m : A_m \rightarrow \mathbb{N}$, which exists since A_m is countable: if it is infinite countable pick the bijection with $\mathbb{N}$ that exists by definition of A_m being infinite countable, if it is finite then just number off its p elements to the first p natural numbers, which is injective. We must now assemble these individual functions into one function $f : A \rightarrow \mathbb{N} \times \mathbb{N}$. Given $x \in A = \bigcup_{i=1}^{\infty} A_i$, pick the unique $m \in M$ for which $x \in A_m$ (because A_m 's are disjoint) and define $f(x) = (m, f_m(x)) \in \mathbb{N} \times \mathbb{N}$. Note that $A_m \neq \emptyset$ since $x \in A_m$, hence $m \in M$ and f_m is well-defined. f is injective because the f_m 's are injective, so if $x, y \in A_j$ then $f(x) \neq f(y)$ by injectivity, and if x, y are in different A_i 's, then our construction of the first coordinate ensures they are not mapped to the same point. Therefore we have injected A into $\mathbb{N} \times \mathbb{N}$, so A is countable since $\mathbb{N} \times \mathbb{N}$ is. ■

(Note: verification of injectivity is mine as an exercise, where a similar exercise could be offered as a bonus to the homeworks in future lectures.)

Proof of Proposition 0.8:

Now take a sequence of (no longer disjoint) countable sets $B_1, \dots, B_n, \dots$. We will use a “disjointing trick”, where we take $A_1 = B_1, A_2 = B_2 \setminus B_1, A_3 = B_2 \setminus (B_1 \cup B_2), \dots, A_n = B_n \setminus (B_1 \cup \dots \cup B_{n-1}), \dots$. Every A_j is countable since $A_j \subseteq B_j$ and B_j is countable, so we can apply Proposition 0.2. The A_j 's are pairwise disjoint by construction, so all we are left with is to prove $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B_n$. To prove the double inclusion, the left side is a subset of the right because $A_j \subseteq B_j$ for all $j \in \mathbb{N}$ (which is preserved under the union), to prove the converse, take an arbitrary $x \in \bigcup_{n=1}^{\infty} B_n$. We want to prove $x \in \bigcup_{n=1}^{\infty} A_n$. Let $P = \{n \in \mathbb{N} \mid x \in B_n\} \subseteq \mathbb{N}$. Then $P \neq \emptyset$ (since $x \in \bigcup_{n=1}^{\infty} B_n$, hence P has a minimal element n_0). If $n_0 = 1$, hence $x \in B_1 = A_1$, then $x \in \bigcup_{n=1}^{\infty} A_n$, if $n_0 > 1$ then $x \in B_{n_0}$, but by minimality of n_0 , we have $x \notin B_1, \dots, B_{n_0-1}$, hence $x \in B_{n_0} \setminus (B_1 \cup \dots \cup B_{n_0-1})$, hence $x \in \bigcup_{n=1}^{\infty} A_n$, proving the other inclusion. Altogether, $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$, where the A_j 's are pairwise disjoint and countable, allowing us to apply the above lemma to get that $\bigcup_{n=1}^{\infty} A_n$ is countable as well, giving that $\bigcup_{n=1}^{\infty} B_n$ is countable. ■

Next time, we will look at Cantor's diagonal trick.

Theorem 0.10:

The interval $[0, 1]$ is not countable.

Proof: Assume by contradiction that someone could write $[0, 1] = \{t_1, t_2, \dots, t_n, \dots\}$ as a sequence. For every $n \in \mathbb{N}$, consider the decimal expansion of $t_n = 0.d_{n1}d_{n2} \dots$ with all d_{ni} 's between 0 and 9 inclusive. If t_n has two decimal expansions, choose the one with 9s. So we have an array of digits of every number in the interval. Now here's Cantor's diagonal trick. Construct a sequence of digits $d_1, d_2, \dots$ such that $d_1 \neq d_{11}, d_2 \neq d_{22}$, in general $d_n \neq d_{nn}$ for all $n \in \mathbb{N}$. For example use the rule that if $d_{nn} \neq 5$ pick 5, otherwise pick 7. Now consider the number $t = 0.d_1d_2 \dots d_n \dots \in [0, 1]$ (we can bound this even tighter than that, between 0.5 and 0.7). But because $d_1 \neq d_{11} \implies |t - t_1| \geq \frac{1}{10^2} \implies t \neq t_1$. Similarly $d_2 \neq d_{22} \implies |t - t_2| \geq \frac{1}{10^3} \implies t \neq t_2$. In general, $d_n \neq d_{nn} \implies |t - t_n| \geq \frac{1}{10^{n+1}} \implies t \neq t_n$. Hence, $t \notin \{t_1, \dots, t_n, \dots\} = [0, 1]$, contradiction, so our assumption was wrong and $[0, 1]$ is not countable. ■

Corollary 0.11:

$\mathbb{R}$ is not countable.

Proof:

By contradiction, assume $\mathbb{R}$ is countable. Then its subsets, including $[0, 1]$, would be countable as well, which we just proved is not the case. ■

Exercise 0.12:

The set $\mathbb{R} \setminus \mathbb{Q}$ of irrational numbers is uncountable.

Definition 1.1:

A metric space (X, d) is a nonempty set X and a function $d : X \times X \rightarrow \mathbb{R}$ satisfying for all $x, y, z \in X$:

- (1) $d(x, y) \geq 0$
- (2) $d(x, y) = 0 \iff x = y$
- (3) $d(x, y) = d(y, x)$
- (4) $d(x, z) \leq d(x, y) + d(y, z)$

Example 1.2:

Take $X = \mathbb{R}^2$, where for $\mathbf{x} = (x^{(1)}, x^{(2)})$, $\mathbf{y} = (y^{(1)}, y^{(2)})$ we have $\sqrt{(x^{(1)} - y^{(1)})^2 + (x^{(2)} - y^{(2)})^2}$. This shows why it's called the triangle inequality.

Example 1.3:

Take $X = \mathbb{R}$, and $d = |x - y|$.

Example 1.4:

Let X be any non-empty set. Define $d : X \times X \rightarrow \mathbb{R}$ by $d(x, y) = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases}$. This is called the discrete

distance. To prove the triangle inequality holds, pick $x, y, z \in X$ and consider the following cases: (1) If $x = y = z$, then $d(x, z) = 0 \leq 0 = 0 + 0 = d(x, y) + d(y, z)$.

(2) Then either $y \neq x$ or $y \neq z$. Then at least one of $d(x, y)$ and $d(y, z)$ are 1, implying $d(x, y) + d(y, z) \geq 1$. But $d(x, z) \leq 1$. ■

More example (1.5 and 1.6)s:

The hamming distance on the hypercube $\{0, 1\}^k$, with $d(x, y) = \sum_{i=1}^k |x^{(i)} - y^{(i)}|$, and generalizes to the weighted Hamming distance $d(x, y) = \sum_{i=1}^k \frac{1}{2^i} |x^{(i)} - y^{(i)}|$ on the space of all infinite strings of 0's and 1's.

Definition 1.7:

Let (X, d) be a metric space, let $a \in X$, and let $r > 0 \in \mathbb{R}$. Denote the open ball $B(a; r) := \{x \in X \mid d(x, a) < r\}$ and the closed ball $\overline{B}(a; r) := \{x \in X \mid d(x, a) \leq r\}$. Look at how these open and closed balls look in various examples.

Proposition 1.9:

- (1) For every $a \in X$ and $r > 0 \in \mathbb{R}$, one has $B(a; r) \subseteq \overline{B}(a; r)$.
- (2) For every $a \in X$ and $r < s \in (0, \infty)$, one has that $\overline{B}(a; r) \subseteq B(a; s)$.
- (3) Let $a \in X$ and $r_1 < r_2 < \dots$ in $(0, \infty)$ such that $\lim_{n \rightarrow \infty} r_n = r \in (0, \infty)$. Then $\bigcup_{n=1}^{\infty} B(a; r_n) = \bigcup_{n=1}^{\infty} \overline{B}(a; r_n) = B(a; r)$.
- (4) Let $a \in X$ and $r_1 < r_2 < \dots$ in $(0, \infty)$ such that $\lim_{n \rightarrow \infty} r_n = \infty$. Then $\bigcup_{n=1}^{\infty} B(a; r_n) = \bigcup_{n=1}^{\infty} \overline{B}(a; r_n) = X$.
- (5) Let $a \in X$ and $r_1 > r_2 > \dots$ in $(0, \infty)$ such that $\lim_{n \rightarrow \infty} r_n = r \in (0, \infty)$. Then $\bigcap_{n=1}^{\infty} B(a; r_n) = \bigcap_{n=1}^{\infty} \overline{B}(a; r_n) = \overline{B}(a; r)$.
- (6) Let $a \in X$ and $r_1 > r_2 > \dots$ in $(0, \infty)$ such that $\lim_{n \rightarrow \infty} r_n = 0$. Then $\bigcap_{n=1}^{\infty} B(a; r_n) = \bigcap_{n=1}^{\infty} \overline{B}(a; r_n) = \{a\}$.

- (7) Let $a \neq b \in X$ and $r, s \in (0, \infty)$ be such that $r + s \leq d(a, b)$. Then $B(a; r) \cap B(b; s) = \emptyset$.
(8) Let $a, b \in X$ and $r, s \in (0, \infty)$ be such that $s \geq r + d(a, b)$. Then $B(a; r) \subseteq B(b; s)$.

Proof:

We prove (2), (3), and (7). Rest are exercises.

(2): Let $x \in \overline{B}(a; r) \implies d(x, a) \leq r < s$. Thus $x \in B(a; s)$.

(3): " $\subseteq$ ": For every $n \in \mathbb{N}$ we have $r_n < r$. Hence $\overline{B}(a; r_n) \subseteq B(a; r)$ (by 2). Hence $\bigcup_{i=1}^{\infty} \overline{B}(a; r_n) \subseteq B(a; r)$. In the reverse inclusion, pick an $x \in B(a; r)$. Because $x \in B(a; r)$, $d(x, a) < r = \lim_{n \rightarrow \infty} r_n$. It then follows that (by definition of the limit) there exists an n_0 such that $r_{n_0} > d(x, a)$. For this n_0 we have $d(x, a) < r_{n_0} \implies x \in B(a; r_{n_0}) \subseteq \overline{B}(a; r_{n_0}) \subseteq \bigcup_{n=1}^{\infty} \overline{B}(a; r_n)$.

(7): Assume for the sake of contradiction that $B(a; r) \cap B(b; s) \neq \emptyset$, hence there exists a $c \in B(a; r) \cap B(b; s)$. Since $c \in B(a; r)$, we have $d(c, a) < r$. Similarly, since $c \in B(b; s)$, we have $d(c, b) < s$. But this tells us that $d(a, b) \leq d(a, c) + d(c, b) < r + s \leq d(a, b)$, contradiction. ■

Proposition 1.11:

Let (X, d) be a metric space, and let $S \subseteq X$. Then TFAE:

- (1) $\exists a \in X$ and $r \in (0, \infty)$ such that $S \subseteq B(a; r)$
(2) $\exists c \geq 0$ such that $d(x, y) \leq c$ for all $x, y \in S$.

Such a set S is called bounded.

Proof:

(1) $\implies$ (2):

Pick $x, y \in S$. Then $x, y \in B(a; r)$, so $d(x, a), d(y, a) < r$, so $d(x, y) \leq d(x, a) + d(a, y) < 2r$. Hence $c = 2r$ works.

(2) $\implies$ (1):

Then there exists $c \geq 0$ such that $d(x, y) \leq c$ for all $x, y \in S$. Pick a point $x_0 \in S$ (assuming S is nonempty, the empty case is vacuously true). Then $d(x, x_0) \leq c < c + 1 \implies x \in B(x_0, c + 1)$ ■

Definition 1.15:

Let $a \in X$. If there exists $r > 0$ such that $B(a; r) = \{a\}$, we say that a is an isolated point of X .

Definition 2.1:

A normed vector space $(X, \|\cdot\|)$ is a vector space X (in this course over $\mathbb{R}$) with a function $\|\cdot\| : X \rightarrow \mathbb{R}$ satisfying for all $x, y \in X$

- (ν1) $\|x\| \geq 0$
(ν2) $\|x\| = 0 \iff x = \mathbf{0}$
(ν3) $\|\alpha x\| = |\alpha| \|x\|$ for $\alpha \in \mathbb{R}$
(ν4) $\|x + y\| \leq \|x\| + \|y\|$

Proposition 2.2:

Every normed vector space $(X, \|\cdot\|)$ is a metric space (X, d) with $d(x, y) = \|x - y\|$.

Proof:

For all $x, y, z \in X$,

(Δ1): $d(x, y) = \|x - y\| \geq 0$ by (ν1).

(Δ2): $d(x, y) = 0 \iff \|x - y\| = 0 \iff x - y = \mathbf{0} \iff x = y$ (second last equivalence by (ν2)).

(Δ3): $d(y, x) = \|y - x\| = \|(-1)(x - y)\| = |-1| \|x - y\| = \|x - y\| = d(x, y)$ where the third equality is (ν3).

(Δ4): $d(x, z) = \|x - z\| \leq \|x - y\| + \|y - z\| = d(x, y) + d(y, z)$, where the inequality is (ν4). ■

Remark 2.3:

Let $(X, \|\cdot\|)$ be a normed vector space. All the terminology from Lecture 1 applies to them under the distance d induced by the norm in the above proposition.

Example 2.4:

$(\mathbb{R}, |\cdot|)$ forms a normed vector space.

Example 2.5:

We can take $\mathbb{R}^k$ with norm $\|\mathbf{x}\| = \sqrt{(x^{(1)})^2 + \cdots + (x^{(k)})^2}$. This norm is called the Euclidean norm (the proof it satisfies the triangle inequality is as seen in linear algebra).

Remark 2.6:

In the above example, the norm is sometimes denoted $\|\cdot\|_2$, because for all $1 \leq p < \infty$ we can define a norm $\|\mathbf{x}\|_p = (\sum_{i=1}^k |x^{(i)}|^p)^{1/p}$. Moreover, there is also a norm for $p = \infty$ given by $\|\mathbf{x}\|_\infty = \max_{i=1}^k |x^{(i)}|$. The norms $p = 1$ and $p = \infty$ respectively induce distances called the “taxicab” and “uniform” distances, denoted by τ and ρ respectively.

Definition 2.7:

Let M be nonempty, and let $f : M \rightarrow \mathbb{R}$ be a function. We say f is bounded if there exists $c \geq 0$ such that for all $x \in M$, we have $|f(x)| \leq c$. We denote by $F_b(M, \mathbb{R})$ the set of all bounded functions from M to $\mathbb{R}$.

Definition 2.8:

We can equip $F_b(M, \mathbb{R})$ with a vector space structure (over $\mathbb{R}$) by pointwise vector addition and scalar multiplication (the zero vector is the identically zero function). Moreover, we can give it a norm, the so-called “infinity norm”:

$$\|f\|_\infty = \sup_{x \in M} |f(x)| \quad \forall f \in F_b(M, \mathbb{R})$$

Proposition 2.9:

Let M be a nonempty set, and consider the vector space $F_b(M, \mathbb{R})$ above. Then $\|\cdot\|_\infty$ is indeed a norm on $F_b(M, \mathbb{R})$.

Proof:

TODO.

We can visualize $F_b(M, \mathbb{R})$ somewhat for intuition: take $M = [0, 1]$, then the space $F_b(M, \mathbb{R})$ is the space of graphs with $[0, 1]$ x -axis without vertical asymptotes. We can think of the uniform distance between two functions in this space to be the length of the longest vertical line segment we can draw between their graphs.

A reason for why $F_b(M, \mathbb{R})$ is useful? “Embedding” of metric spaces inside normed vector spaces:

Example:

Let $X = \{0, 1\}^3$, so $|X| = 8$. On this X , consider the Hamming distance metric. But notice that for this case, $d_H(x, y) = \sum_{k=1}^3 |x^{(k)} - y^{(k)}|$, if we let x and y vary beyond having 0s and 1s (any real numbers), we just have $\mathbb{R}^3$, and our metric is the distance induced by $\|\cdot\|_1$, so we can consider (X, d_H) as a subspace of the normed vector space $(\mathbb{R}^3, \|\cdot\|_1)$.

Question:

More generally, let (X, d_X) be a metric space. Is it always true that we can find a normed vector space $(Y, \|\cdot\|)$ and an injective function $\phi : X \rightarrow Y$ such that $d_X(x_1, x_2) = \|\phi(x_1) - \phi(x_2)\|$?

Hint:

The answer is yes, and we can use $Y = F_b(X, \mathbb{R})$.

Definition/Proposition 3.2:

Let (X, d) be a metric space, let $(x_n)_{n=1}^\infty$ be a sequence in X , with $a \in X$. Then $(x_n)_{n=1}^\infty$ is said to converge to a if and only if for all $\varepsilon > 0$, there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $d(x_n, a) < \varepsilon$ if and only if $\lim_{n \rightarrow \infty} d(x_n, a) = 0$ in the sense of calculus if and only if for all $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that $x_n \in B(a; \varepsilon)$ for all $n \geq n_0$.

Proposition 3.6:

Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$. Then $a = b$.

Proof:

Assume by contradiction that $a \neq b$. Let $r = d(a, b) > 0$. We have $B(a; \frac{r}{3}) \cap B(b; \frac{r}{3}) = \emptyset$ (by Proposition 1.9(7)). Since $x_n \rightarrow a$, can find $n_0 \in \mathbb{N}$ such that $x \in B(a; \frac{r}{3})$ for all $n \geq n_0$. Since $x_n \rightarrow b$, can find n'_0 such that $x_n \in B(b; \frac{r}{3})$ for all $n \geq n'_0$. Take $n = \max(n_0, n'_0)$. Then $x_n \in B(a; \frac{r}{3})$ and $x_n \in B(b; \frac{r}{3})$, hence $x_n \in B(a; \frac{r}{3}) \cap B(b; \frac{r}{3}) = \emptyset$, contradiction, hence $a = b$. ■

Corollary 3.7:

In a metric space (X, d) instead of $x_n \rightarrow a$, we can also write $a = \lim_{n \rightarrow \infty} x_n$ because the limit is uniquely determined. ■

Definition 3.8:

Let $(x_n)_{n=1}^\infty$ be a sequence in (X, d) with $a \in X$. a is a cluster point for $(x_n)_{n=1}^\infty$ if and only if for all $\varepsilon > 0$ the sequence $(x_n)_{n=1}^\infty$ visits the ball $B(a; \varepsilon)$ infinitely many times (i.e. for every $n_0 \in \mathbb{N}$, can find $n \geq n_0$ such that $x_n \in B(a; \varepsilon)$).

Proposition 3.11:

Let (X, d) be a metric space. Let $(x_n)_{n=1}^\infty$, with $a \in X$. Then a is a cluster point (as in the above definition) if and only if there exists a subsequence $(x_{n(k)})_{k=1}^\infty$ that converges to a .

Proof:

($\implies$): We know that a is a cluster point of $(x_n)_{n=1}^\infty$. Hence there exists $n(1)$ such that $x_{n(1)} \in B(a; 1)$. Similarly there exists $n(2) > n(1)$ such that $x_{n(2)} \in B(a; \frac{1}{2})$, and inductively $n(k) > n(k-1)$ such that $x_{n(k)} \in B(a; \frac{1}{k})$. This gives us our subsequence $1 \leq n(1) \leq n(2) < \dots < n(k) < \dots$ and $0 \leq d(x_{n(k)}, a) < \frac{1}{k}$ for all $k \in \mathbb{N}$. Hence $d(x_{n(k)}, a) \rightarrow 0$ by the squeeze theorem. Hence $x_{n(k)} \rightarrow a$.

($\impliedby$): Easy exercise. ■

Corollary 3.12:

Let (X, d) be a metric space, and let $(x_n)_{n=1}^\infty$ be a sequence in X . If $(x_n)_{n=1}^\infty$ is convergent, then it has one cluster point, namely its limit.

Proof:

This holds because every subsequence of $(x_n)_{n=1}^\infty$ converges to the same limit as the whole sequence, so by the above equivalence, every cluster point must be the limit of the whole sequence. ■

Example 3.13:

The converse is not true, take the sequence $(0, 1, 0, 2, 0, 3, 0, 4, \dots)$, it has a unique cluster point 0 but does not converge.

Definition 3.14:

We say a sequence $(x_n)_{n=1}^\infty$ is bounded when $d(x_m, x_k) \leq c$ for some $c \in \mathbb{R}$ and all $m, k \in \mathbb{N}$.

Exercise 3.15:

The above example uses an unbounded sequence. Can we find a metric space (X, d) and a bounded sequence $(x_n)_{n=1}^\infty$ such that $(x_n)_{n=1}^\infty$ has a unique cluster point but does not converge?

Take $X = \mathbb{R}$, but d to be the discrete distance. Then take $(x_n)_{n=1}^\infty = (0, 1, 0, 2, 0, 3, \dots, 0, n, \dots)$. This is bounded (any sequence in this metric space is bounded), but it still does not converge, but still has a cluster point $a = 0$. This uses the fact that convergent sequences in the discrete distance are those that are eventually constant.

Definition 4.1:

Let d and d' be two distance functions on the same set X . We say d and d' are equivalent distances if

they have exactly the same convergent sequences with the same limits.

Definition 4.2:

A stronger condition than the above: Let X be nonempty, and let d and d' be two distance functions on X . We will say that d and d' satisfy the bi-Lipschitz condition if there exist $\alpha, \beta > 0$ such that

$$\alpha \leq \frac{d(x_1, x_2)}{d'(x_1, x_2)} \leq \beta \quad \forall x_1 \neq x_2 \in X$$

Proposition 4.3:

If d and d' are bi-Lipschitz, then d and d' are equivalent.

Proof:

Suppose $(x_n)_{n=1}^\infty$ is a sequence in X , and let $a \in X$. We must show $x_n \xrightarrow{d} a$ if and only if $x_n \xrightarrow{d'} a$. We show the forward direction (the reverse is identical with the roles of d, d' reversed). So assume that $x_n \xrightarrow{d} a$. For every $n \geq 1$, the bi-Lipschitz condition gives that $d'(x_n, a) \leq \frac{1}{\alpha} d(x_n, a)$ so then $0 \leq d'(x_n, a) \leq \frac{1}{\alpha} d(x_n, a)$. But $d(x_n, a) \rightarrow 0$ by convergence of $(x_n)_{n=1}^\infty$, so $d'(x_n, a) \rightarrow 0$ by the squeeze theorem (since multiplying by the constant $\frac{1}{\alpha}$ does not change anything), which exactly means that $(x_n)_{n=1}^\infty$ converges to a under the distance d' . ■

Example:

On $X = \mathbb{R}^k$ we have norms $\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_\infty$, giving rise to the taxicab distance τ , the Euclidean distance d , and the distance we denoted ρ respectively. Any two of τ, d, ρ satisfy bi-Lipschitz, hence are equivalent, which comes from the fact that $\|x\|_\infty \leq \|x\|_p \leq k\|x\|_\infty$, where $k = \dim X$. An example of this explicitly: For every $x_1, x_2 \in \mathbb{R}^k$, we have $\rho(x_1, x_2) = \|x_1 - x_2\|_\infty \leq \|x_1 - x_2\|_1 = \tau(x_1, x_2)$, and $\tau(x_1, x_2) = \|x_1 - x_2\|_1 \leq k\|x_1 - x_2\|_\infty = k\rho(x_1, x_2)$. Hence we have $1 \leq \frac{\tau(x_1, x_2)}{\rho(x_1, x_2)} \leq k$.

Remark 4.4 - quiz:

Give an example of a set X and two distance functions d, d' on X that are equivalent but not bi-Lipschitz.

We want to find a special situation when the converse of Proposition 4.3 does hold:

Proposition 4.5:

Let X be a real vector space. Suppose $\|\cdot\|$ and $\|\cdot\|'$ are norms on X , with associated distances d and d' . Then TFAE:

- (1) d and d' are equivalent distances.
- (2) d and d' are bi-Lipschitz.
- (3) There exist $\alpha > 0, \beta > 0$ such that for any $0_X \neq x \in X$, then $\alpha \leq \frac{\|x\|}{\|x\|'} \leq \beta$.

Proof:

We prove (3) $\implies$ (2) $\implies$ (1) $\implies$ (3). (2) $\implies$ (1) is Proposition 4.3.

Assume (3). Then, there exist $\alpha > 0$ and $\beta > 0$ such that $\alpha \leq \frac{\|x\|}{\|x\|'} \leq \beta$. We claim these same constants work for bi-Lipschitz, indeed, for $x \neq y \in X$ we have

$$\frac{d(x, y)}{d'(x, y)} = \frac{\|x - y\|}{\|x - y\|'} \in [\alpha, \beta]$$

by applying (3) to the vector $x - y \in X$ ($x \neq y$ assures us $x - y \neq 0_X$), and we are done.

We now prove (1) $\implies$ (3). Assume (1), so d and d' are equivalent. Assume by contradiction that there is no $\alpha > 0$ (the case for β is analogous with $\beta = \frac{1}{\alpha}$) such that $\frac{\|x\|}{\|x\|'} \geq \alpha$ for all nonzero $x \in X$. This means for every $n \in \mathbb{N}$, we have $\frac{\|x_n\|}{\|x_n\|'} < \frac{1}{n}$ for some nonzero $x_n \in X$ depending on n . For every $n \in \mathbb{N}$ denote $\|x_n\| = p_n$ and $\|x_n\|' = q_n$, so $p_n, q_n > 0$ with $\frac{p_n}{q_n} < \frac{1}{n}$. For every $n \in \mathbb{N}$, denote $y_n = \frac{1}{q_n} x_n \in X$. This gives a sequence $(y_n)_{n=1}^\infty$.

Claim 1: $d(y_n, 0_X) < \frac{1}{n}$ for all $n \in \mathbb{N}$ and therefore $y_n \xrightarrow{d} 0_X$.

Claim 2: $d'(y_n, 0_X) = 1$ for all $n \in \mathbb{N}$, and therefore y_n does not converge in d' to 0_X .

Once we prove these claims, we will have our contradiction because d and d' are equivalent.

Proving claim 1, $d(y_n, 0_X) = \|y_n\| = \|\frac{1}{q_n}x_n\| = \frac{\|x_n\|}{q_n} = \frac{p_n}{q_n} < \frac{1}{n}$. Thus, since $0 \leq d(y_n, 0_X) < \frac{1}{n} \rightarrow 0$, by the squeeze theorem we have $d(y_n, 0_X) \rightarrow 0$ so $y_n \xrightarrow{d} 0_X$.

Proving claim 2, $d'(y_n, 0_X) = \|y_n\|' = \frac{\|x_n\|'}{q_n} = \frac{q_n}{q_n} = 1$. Therefore $d'(y_n, 0_X)$ does not converge to 0 under d' .

Therefore, d and d' are not equivalent, contradiction, so such an $\alpha > 0$ such that $\frac{\|x\|}{\|x\|'} \geq \alpha$ must exist, and then choose $\beta = \frac{1}{\alpha}$. ■

Definition 4.6:

Let X be a vector space over $\mathbb{R}$, with two norms $\|\cdot\|$ and $\|\cdot\|'$. We define condition (3) above to be the notion that $\|\cdot\|$ and $\|\cdot\|'$ are equivalent norms.

Example 4.7:

On $\mathbb{R}^k$, the norms $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$ are equivalent to each other, this comes from the inequalities (for all $x \neq \mathbf{0} \in \mathbb{R}^k$)

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq k\|x\|_\infty$$

where $k = \dim \mathbb{R}^k$.

Definition 4.8:

Define $\mathbb{R}_{fin}^\infty$ to be the vector space over $\mathbb{R}$ of all sequences in $\mathbb{R}$ that are eventually 0.

Remark 4.9:

This is basically what happens when we take $\mathbb{R}^k$ for all $k \in \mathbb{N}$, we embed them into each other by appending a 0 in coordinates, and then take their infinite union. Defining the norms $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$ naturally, we see they are NOT equivalent on $\mathbb{R}_{fin}^\infty$.

Definition 5.1:

Let (X, d) and (Y, d') be two metric spaces, and let $f : X \rightarrow Y$ be a function. We say that f is continuous when it respects convergent sequences: given $(x_n)_{n=1}^\infty$ such that $\lim_{n \rightarrow \infty} x_n = x$, we have $\lim_{n \rightarrow \infty} f(x_n) = f(x)$.

Definition 5.7:

Take the definition as above, and pick $a \in X$. Then we say f is continuous at a if for any convergent sequence $(x_n)_{n=1}^\infty$ such that $\lim_{n \rightarrow \infty} x_n = a$, we have $\lim_{n \rightarrow \infty} f(x_n) = f(a)$. We see f is continuous on X if and only if it is continuous at every $a \in X$. This definition is equivalent to the infamous $\varepsilon - \delta$ condition: For every $\varepsilon > 0$, there exists $\delta > 0$ such that $x \in X, d(a, x) < \delta \implies d'(f(a), f(x)) < \varepsilon$, with the exact same proof as Calculus 1.

Let's look at compositions and inverses of continuous functions.

Remark 5.2:

Given $(X, d), (Y, d'), (Z, d'')$, we see that if f and g are continuous, then $g \circ f : X \rightarrow Z$ is continuous.

Proof:

Pick x and $(x_n)_{n=1}^\infty$ in X such that $x_n \xrightarrow{d} x$. Then $f(x_n) \xrightarrow{d'} f(x)$ by continuity. Hence $g(f(x_n)) \xrightarrow{d''} g(f(x))$ by g 's continuity, so $g \circ f$ respects convergent sequences and thus is continuous. ■

What about inverses?

Definition 5.3:

Let $f : X \rightarrow Y$ be a bijective function from (X, d) to (Y, d) . We say that f is a homeomorphism if both

f and f^{-1} are continuous.

Exercise 5.4:

There is a reason why we didn't define a homeomorphism as a continuous bijection, it's inverse need not be continuous: Take $X = Y = \mathbb{R}$, $f = \text{id}$, and equip X with the discrete distance, Y with the usual distance. f is obviously bijective, and take a sequence $(x_n)_{n=1}^\infty$ that converges to some x in the discrete distance, i.e. $(x_n)_{n=1}^\infty$ is eventually constant. Eventually constant sequences also converge in the usual distance, so $f(x_n) \xrightarrow{d'} f(x)$, so f is continuous. Conversely, a convergent sequence in the usual distance need not be eventually constant, so $f^{-1}(x_n) \not\xrightarrow{d} f(x)$ for some convergent sequence $(x_n)_{n=1}^\infty$ in (Y, d') . For another example, see Problem P5.2.

Exercise 5.6:

Take $X = (\mathbb{R}, d)$ with the usual distance, $Y = ((0, 1), d)$ with the restriction of the usual distance. Prove that (X, d) and (Y, d') are homeomorphic.

Proof:

Define $g : (0, 1) \rightarrow \mathbb{R}$ by $g(y) = \tan(\pi(y - \frac{1}{2}))$. To verify it is bijective, we come up with its inverse $f = g^{-1} : \mathbb{R} \rightarrow (0, 1)$, which can be seen to be $f(x) = \frac{\arctan(x)}{\pi} + \frac{1}{2}$. These are both continuous (via calculus). Therefore g is a homeomorphism, since it is a continuous bijection with continuous inverse. Thus $(0, 1)$ and $\mathbb{R}$ are homeomorphic. Notably, we see boundedness of $(0, 1)$ is not preserved.

Definition 5.9:

Let (X, d) and (Y, d') be metric spaces, with $f : X \rightarrow Y$. Let $c \geq 0$. We say f is c -Lipschitz if we have that $d'(f(x_1), f(x_2)) \leq cd(x_1, x_2)$ for all $x_1, x_2 \in X$. If f is 1-Lipschitz, we call it a contraction, and if f is c -Lipschitz for $c \in [0, 1)$ we say it is a strict contraction. A function that is c -Lipschitz for some c is just called Lipschitz.

Remark 5.10:

Every Lipschitz function is continuous. The converse does not hold.

Proof:

Suppose (X, d) and (Y, d') are Lipschitz, with $f : X \rightarrow Y$. If f is Lipschitz, then f is continuous: So consider x and $(x_n)_{n=1}^\infty$, where $x_n \xrightarrow{d} x$, so $d(x_n, x) \rightarrow 0$. But $0 \leq d'(f(x_n), f(x)) \leq cd(x_n, x) \rightarrow 0$, so by the squeeze theorem we have $d'(f(x_n), f(x)) \rightarrow 0$. ■

The converse does not hold. Take $X = Y = [1, \infty)$, where $d = d'$ is the restriction of the usual distance. Let $f : X \rightarrow Y$ be $f(x) = \sqrt{x}$. f is 1/2-Lipschitz, indeed, let $x_1, x_2 \in [1, \infty)$. Then $d'(f(x_1), f(x_2)) = |f(x_1) - f(x_2)| = |\sqrt{x_1} - \sqrt{x_2}| = \left| \frac{x_1 - x_2}{\sqrt{x_1} + \sqrt{x_2}} \right| = \frac{|x_1 - x_2|}{\sqrt{x_1} + \sqrt{x_2}} \leq \frac{|x_1 - x_2|}{2} = \frac{1}{2}d(x_1, x_2)$. On the other hand, let $g : Y \rightarrow X$, $g(y) = y^2$ for all $y \in Y$, note that f and g are bijective with $g = f^{-1}$. This g is not Lipschitz. Why is it that for no $c \in [0, \infty)$ we have $|y_1^2 - y_2^2| \leq c|y_1 - y_2|$ (Exercise). So f is an example of a bijective Lipschitz function, but its inverse g is not Lipschitz but is clearly continuous.

Remark 5.11:

Lipschitz functions are closed under composition (if f is c_1 -Lipschitz and g is c_2 -Lipschitz then $f \circ g$ is at most c_1c_2 -Lipschitz), but not necessarily closed under inverses (see above).

What happens if we look at Lipschitz functions on normed vector spaces? Something really cool happens:

Theorem 5.13:

Let $(X, \|\cdot\|)$ and $(Y, \|\cdot\|')$ be two normed vector spaces, and let $f : X \rightarrow Y$ be linear. TFAE:

- (1) There exists $x \in X$ such that f is continuous at x .
- (2) f is continuous at 0_X .

- (3) f is continuous.
- (4) f is Lipschitz.
- (5) f is bounded on the closed unit ball centered at the origin.

Proof:

The fact (3) $\implies$ (2) $\implies$ (1) is satisfied trivially, and every Lipschitz function is continuous, so (4) $\implies$ (3).

We next prove (5) $\implies$ (4). Then there exists $c > 0$ such that $\|f(x)\|' \leq c$ for all $x \in X$ satisfying $\|x\| \leq 1$. But because f is linear, for all $x_1, x_2 \in X$,

$$\|f(x_1) - f(x_2)\|' = \|f(x_1 - x_2)\|'$$

so it suffices to find $b > 0$ such that $\|f(v)\|' \leq b\|v\|$ for all $v \in X$. We claim the c we got from (5) works for our choice of b . If $v = 0_X$, our claim holds trivially. So suppose $v \neq 0_X$. Hence $\|v\| > 0$, and we consider $u = \frac{1}{\|v\|}v \in X$, so we get $\|u\| = 1$. Hence our above estimate on the unit ball applies, and gives $\|f(u)\|' \leq c$. But $c \geq \|f(u)\|' = \|f(v/\|v\|)\|' = \|f(v)\|'/\|v\|$ (by linearity of f and the norm), which after multiplying by $\|v\|$, is exactly what we wanted. This completes the proof of this implication.

We finish by proving (1) $\implies$ (5). Therefore, there exists $x_o \in X$ such that f is continuous at x_o . By continuity, for all $\varepsilon > 0$ there exists $\delta > 0$ such that for $x \in X$ satisfying $\|x - x_o\| < \delta$, then $\|f(x) - f(x_o)\|' < \varepsilon$. In particular, choose $\varepsilon = 1$ and let δ denote what we get from this. We claim if $v \in X$, with $\|v\| \leq 1$, then $\|f(v)\|' \leq 2/\delta$, which will complete the proof since v is in our unit ball (and we will have found a bound for it). Suppose we have $\|v\| \leq 1$. Define $x = x_o + \frac{\delta}{2}v$. Then, $\|x - x_o\| = \|\frac{\delta}{2}v\| \leq \frac{\delta}{2} < \delta$. Then, by continuity, we have $\|f(x) - f(x_o)\|' < 1$. But $\|f(x) - f(x_o)\|' = \|f(x - x_o)\|' = \|f(\frac{\delta}{2}v)\|' = \frac{\delta}{2}\|f(v)\|'$ by linearity of f and the norm. Therefore, we have $\frac{\delta}{2}\|f(v)\|' < 1$, so $\|f(v)\|' < \frac{2}{\delta}$, completing the proof of our claim and thus this implication. ■

Comment:

Suppose that $(X, \|\cdot\|)$ and $(Y, \|\cdot\|')$ are normed vector spaces, $f : X \rightarrow Y$ linear. It may happen that f is not continuous, which the above theorem shows is *strongly* not continuous - not even at a single point. See P5.5 and P5.6.

Definition 6.1:

Let (X, d) be a metric space, with $A \subseteq X$. Then

(1) An element $a \in A$ is an interior point of A when there exists $r > 0$ such that $B(a; r) \subseteq A$. The set of interior points of A is called its interior $\text{Int } A$ or A° .

(2) An element $b \in X$ is said to be adherent to A when it has the property that $B(b; r) \cap A \neq \emptyset$ for all $r > 0$. The set of all adherent points of A is called the closure of A and is denoted $\text{Cl } A = \bar{A}$.

We have $\text{Int } A \subseteq A \subseteq \text{Cl } A$. Indeed, if $a \in A$, then $a \in B(a; r)$ for all $r > 0$, so $B(a; r) \cap A \neq \emptyset$ (it contains a), so a is adherent, so any point of A is adherent so it is in the closure. A similar obvious verification shows the other inclusion.

Definition 6.2:

Define $\text{Bd } A$ or ∂A (called the boundary) to be $\text{Cl } A \setminus \text{Int } A$.

Exercise 6.3:

Let $A = \{(s, t) \mid s, t \in \mathbb{R}, t > 0\} \cup \{(s, 0) \mid s \geq 0\}$. Then $\text{Int}(A)$ is the upper-half plane, $\text{Cl}(A)$ adds the rest of the x -axis, $\text{Bd}(A)$ is just the x -axis.

Tonight's quiz: Prove why for every subset in the discrete distance, $\text{Cl}(A) = A$.

Proposition 6.4:

$\text{Cl } A = \{b \in X \mid \text{there exists } (x_n)_{n=1}^\infty \text{ in } X \text{ that converges to } b\}$

Proof:

We prove both inclusions:

$\subseteq$:

Pick $b \in \text{Cl } A$, then for all $r > 0$, $B(b; r) \cap A \neq \emptyset$. We want a sequence $(x_n)_{n=1}^\infty$ in A that converges to b , but for every $n \in \mathbb{N}$, we can set $r = \frac{1}{n}$ and find a point $x_n \in B(b; r) \cap A$, which gives us our sequence. For every $n \in \mathbb{N}$, we have $d(b, x_n) < \frac{1}{n} \rightarrow 0$, so by the squeeze theorem we have $(x_n)_{n=1}^\infty \rightarrow b$, completing the proof of this direction.

$\supseteq$:

Pick $b \in X$ that is a limit of some sequence $(x_n)_{n=1}^\infty$ in A . Pick $r > 0$ for which we will verify that $B(b; r) \cap A \neq \emptyset$. By convergence with $\varepsilon = r$, there exists $n_o \in \mathbb{N}$ such that $d(x_n, b) < r$ for all $n \geq n_o$. Then clearly, we have $x_{n_o} \in B(b; r) \cap A$, since all x_n 's are in A and $d(x_{n_o}, b) < r$. ■

Proposition 6.5 (“duality” between closure and interior):

We have $X \setminus \text{Cl } A = \text{Int}(X \setminus A)$, and $X \setminus \text{Int } A = \text{Cl}(X \setminus A)$. This is analogous to DeMorgan’s laws in set theory: $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$, and $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$.

Proof:

We will do the first equality (proof of second equality is similar).

$\subseteq$:

Pick $b \in X \setminus \text{Cl } A$. So $b \in X$, and $b \notin \text{Cl } A$, so there exists $r > 0$ such that $B(b; r) \cap A = \emptyset$. For this $r > 0$, we have that $B(b; r) \subseteq X \setminus A$ (since $A \subseteq \text{Cl } A$). But this means by definition that $b \in \text{Int}(X \setminus A)$, completing the proof of this inclusion.

$\supseteq$:

Reverse the above argument. ■

Remark 6.7:

It is immediate that $M \subseteq N$ implies $\text{Int } M \subseteq \text{Int } N$, and $\text{Cl } M \subseteq \text{Cl } N$.

Proposition 6.8:

- (1) $\text{Int}(A_1 \cap \cdots \cap A_n) = \text{Int } A_1 \cap \cdots \cap \text{Int } A_n$ and
- (2) $\text{Int}(A_1 \cup \cdots \cup A_n) \supseteq \text{Int } A_1 \cup \cdots \cup \text{Int } A_n$.

Dual statements hold for the closure.

Proof:

For $\subseteq$ of (1), let $A_1 \cap \cdots \cap A_n = M$. Then $M \subseteq A_i$ for all $i = 1, \dots, n$, so $\text{Int } M \subseteq \text{Int } A_i$ by the above remark, so $\text{Int } M \subseteq \text{Int } A_1 \cap \cdots \cap \text{Int } A_n$. The proof of (2) is similar (note that the inclusion is strict, where we take $A_1 = \mathbb{Q}$ and $A_2 = \mathbb{R} \setminus \mathbb{Q}$ as an example of why).

For $\supseteq$ of (1), pick a point $a \in \text{Int}(A_1) \cap \cdots \cap \text{Int}(A_n)$. Then, $a \in \text{Int}(A_i)$ for all $i \in 1, \dots, n$, so $\exists r_1 > 0$ such that $B(a; r_1) \subseteq A_1$. Let $r = \min\{r_1, \dots, r_n\} \in (0, \infty)$, so $r \leq r_i \implies B(a; r) \subseteq B(a; r_i) \subseteq A_i$, so $B(a; r) \subseteq A_1 \cap \cdots \cap A_n$, so a is an interior point of $A_1 \cap \cdots \cap A_n$, so $\text{Int}(A_1) \cap \cdots \cap \text{Int}(A_n) \subseteq \text{Int}(A_1 \cap \cdots \cap A_n)$, proving this inclusion. ■

Definition 7.1:

Let (X, d) be a metric space, and let $D \subseteq X$. Then D is an open set if and only if $\text{Int}(D) = D$ (occurring iff $D \subseteq \text{Int}(D)$) if and only if for every $a \in D$, there exists $r > 0$ such that $B(a; r) \subseteq D$.

Notation:

The collection $\tau := \{D \subseteq X \mid D \text{ open}\} \subseteq 2^X$ (2^X is the powerset of X) is called the topology of (X, d) .

Proposition 7.2:

Let (X, d) be a metric space, where τ is its topology. Then:

- (1): $\emptyset \in \tau$
- (2): $X \in \tau$
- (3): If $(D_i)_{i \in I}$ are in τ (for some index set I), then $\bigcup_{i \in I} D_i \in \tau$
- (4): If $D_1, \dots, D_n \in \tau$, then $D_1 \cap \cdots \cap D_n \in \tau$.

Proof:

For (1), vacuously $\emptyset \subseteq \text{Int}(\emptyset)$ and $\text{Int}(\emptyset) \subseteq \emptyset$, so $\emptyset = \text{Int}(\emptyset)$ so $\emptyset$ is open.

(2) For every $a \in X$, we have $B(a; 1) \subseteq X$, so $a \in \text{Int}(X)$ so $X \subseteq \text{Int}(X)$, so X is open.

(3) Let $x \in \bigcup_{i \in I} D_i$. Then there is i_o such that $x \in D_{i_o}$, where since that set is open, there is $\varepsilon > 0$ s.t. $B(x; \varepsilon) \subseteq D_{i_o} \subseteq \bigcup_{i \in I} D_i$, so x is interior to that union and thus the union is open.

(4) Let $x \in D_1 \cap \dots \cap D_n$. Then x is in each D_i , and so there exist $r_1, \dots, r_n > 0$ such that $B(x; r_i) \subseteq D_i$. Take $r = \min\{r_i \mid i \in \{1, \dots, n\}\}$, giving $B(x; r) \subseteq D_1 \cap \dots \cap D_n$. ■

Proposition 7.4:

Let (X, d) be a metric space, and let $x \in X$ and $r > 0$. Then $B(x; r)$ is an open subset of X .

Proof:

TODO.

Lemma 7.5/Lemma 7.6/Proposition 7.7:

Let (X, d) be a metric space, and let A be a subset of X . Then $\text{Int } A$ is the largest open subset of A contained in A . That is,

(i, Lemma 7.6) $\text{Int } A$ is open and $\text{Int } A \subseteq A$

(ii, Lemma 7.5) if D is any open set such that $D \subseteq A$, then $D \subseteq \text{Int } A$.

Proof:

TODO.

Definition 7.8:

A topological space $(X, \mathcal{T})$ where X is a non-empty set and $\mathcal{T}$ is a collection of subsets of X satisfying:

(i) $X \in \mathcal{T}$

(ii) $\emptyset \in \mathcal{T}$

(iii) If $(D_i)_{i \in I} \subseteq \mathcal{T}$, then $\bigcup_{i \in I} D_i \in \mathcal{T}$

(iv) If $D_1, \dots, D_n \in \mathcal{T}$, then $D_1 \cap \dots \cap D_n \in \mathcal{T}$.

Proposition 7.2 shows that a metric space and its topology make it into a topological space. Moreover, we have something stronger:

Definition 7.9:

Let $(X, \mathcal{T})$ be a topological space. We say that $(X, \mathcal{T})$ is Hausdorff when for any $a, b \in X$ such that $a \neq b$, there exist $U, V \in \mathcal{T}$ with $a \in U$, $b \in V$, and $U \cap V = \emptyset$.

Proposition 7.10:

Let (X, d) be a metric space with τ its topology. Then (X, τ) is a Hausdorff topological space.

Proof:

Let $a \neq b \in X$. Then $r := d(a, b)/3$ works to give open balls $B(a; r)$ and $B(b; r)$ that separate the points. ■

Definition/Proposition 8.1:

Let (X, d) be a metric space and let $F \subseteq X$. TFAE:

(1) $F = \text{cl } F$

(2) The complement $D := X \setminus F$ is an open subset of X .

(3) The all convergent sequences in F have limits that are in F .

Such a set F meeting any of the three (and thus all of the three) properties is called a closed subset of X .

Proposition 8.3:

Let (X, d) be a metric space. Then,

(1) $\emptyset$ is closed

(2) X is closed

- (3) The finite union of closed sets is closed
- (4) The arbitrary intersection of closed sets is closed

Proof:

Apply Proposition 7.2, Definition 8.1(2), and DeMorgan's Law. ■

Proposition 8.5:

Let (X, d) be a metric space, let a be a point of X , and let $r > 0$ be a real number. Then the closed ball $\overline{B}(a; r)$ is closed.

Proposition 8.6:

Let (X, d) be a metric space, and let $A \subseteq X$. Then $\text{cl}(A)$ is the smallest closed set containing A .

Definition 9.1:

Let (X, d) be a metric space, and let A be a nonempty subset of X , and let us moreover consider a subset S of A . The following three statements about S and A are equivalent to each other:

- (1) For every $a \in A$, there exists a sequence $(x_n)_{n=1}^\infty$ such that $\lim_{n \rightarrow \infty} x_n = a$.
- (2) $\text{cl}(S) \supseteq A$.
- (3) $\text{cl}(S) = \text{cl}(A)$.

Proof of equivalence:

The description of $\text{cl}(S)$ in terms of sequences is exactly (1), giving $(1) \iff (2)$. $(3) \implies (2)$ is clear, since if $\text{cl}(S) = \text{cl}(A)$, we have $A \subseteq \text{cl}(A) = \text{cl}(S)$. (2) implies (3) follows since $\text{cl}(A) \subseteq \text{cl}(S)$ by minimality of the closure of A , since $\text{cl}(S)$ is a closed set containing A . The other inclusion follows since $S \subseteq A$, so $\text{cl}(S) \subseteq \text{cl}(A)$. ■

The prototypical example from Calculus 1 is $X = \mathbb{R}$, d to be the usual distance, take $A = [0, 1]$, and $S = [0, 1] \cap \mathbb{Q}$ is dense in A .

Definition 9.3:

Let (X, d) be a metric space, with $A \subseteq X$. We say that A is separable if there exists $S \subseteq A$ that is countable and dense in A . In the special case where $A = X$, we say X is separable if the above definition applies, i.e. X has a countable subset S such that $\text{cl}(S) = X$.

Remark 9.4:

The condition $\text{cl}(S) = X$ can be viewed as “ S meets all the open balls in the space X ”.

Example 9.5:

- (1) $X = \mathbb{R}$ with the usual distance is separable.
- (2) $X = \mathbb{R}^2$ with the Euclidean distance is separable.
- (3) If we take X to be any uncountable set with the discrete distance, it is not separable since $\text{cl}(A) = A$ for all sets A .

Definition 9.6:

Let (X, d) be a metric space and let $\mathcal{T}$ be X 's topology. A collection $\mathcal{G} \subseteq \mathcal{T}$ of open sets is said to be a basis of open sets for (X, d) , whenever $D \in \mathcal{T}$ and $a \in D$, there exists $G \in \mathcal{G}$ such that $a \in G \subseteq D$.

This is equivalent to the statement that for all nonempty $D \in \mathcal{T}$, we can find a family $(G_i)_{i \in I}$ of sets from $\mathcal{G}$ such that $\bigcup_{i \in I} G_i = D$.

Proof:

Assume the statement given in the definition. For every $a \in D$, we can find $G_a \in \mathcal{G}$ such that $a \in G_a \subseteq D$. Then, we claim $D = \bigcup_{a \in D} G_a$. The forward inclusion is by construction. In the reverse inclusion, every G_i is a subset of D , so their union is at most D .

Conversely, given $D \in \mathcal{T}$ and $a \in D$, we can write $D = \bigcup_{i \in I} G_i$ by hypothesis for some family $(G_i)_{i \in I}$ in $\mathcal{G}$. Then $a \in \bigcup_{i \in I} G_i$, so it must in one of the sets we are taking the union of, G_{i_0} , hence $a \in G_{i_0}$ so

$$a \in G_{i_o} \subseteq \bigcup_{i \in I} G_i = D. \blacksquare$$

Definition 9.9:

(X, d) is said to be second countable when there is a basis $\mathcal{G}$ of open sets for (X, d) such that $\mathcal{G}$ is countable.

Theorem 9.10:

(X, d) is separable if and only if it is second countable.

Proof (might partially be on the final):

($\Leftarrow$):

We have that there exists a basis $\mathcal{G}$ of open sets that is countable. WLOG assume $\mathcal{G} = \{G_1, \dots, G_n, \dots\}$ is infinite countable with all sets nonempty. For every $n \in \mathbb{N}$, pick a point $x_n \in G_n$. Let $S = \{x \in X \mid x_n = x \text{ for some } n \in \mathbb{N}\}$. We have a surjective map $\phi : \mathbb{N} \rightarrow S$ by $\phi(n) = x_n$, so S is countable (by the surjective tool). We next claim S is dense in X . By Remark 9.4, we will equivalently show that $S \cap B(a; r) \neq \emptyset$ for all $a \in X$ and $r > 0$. By definition of being a basis, we can find $n \in \mathbb{N}$ such that $a \in G_n \subseteq B(a; r)$. We have $x_n \in G_n$, so $x_n \in B(a; r) \cap S$ since $x_n \in S$, so $S \cap B(a; r) \neq \emptyset$. This completes the proof of this direction, S is separable.

($\Rightarrow$):

We have that there exists $S \subseteq X$ countable and dense. WLOG suppose S is infinite countable (otherwise X is finite and trivially second countable), write $S = \{x_1, \dots, x_n, \dots\}$. Take $\mathcal{G} = \{G \subseteq X \mid \exists n, m \in \mathbb{N} \text{ s.t. } G = B(x_n; \frac{1}{m})\}$. This is a collection of open sets, since every open ball is open. We first claim that this is countable. Indeed, let $\psi : \mathbb{N} \times \mathbb{N} \rightarrow \mathcal{G}$ be given by $\psi(n, m) = B(x_n; \frac{1}{m})$. This is surjective by construction, and $\mathbb{N} \times \mathbb{N}$ is countable so by the surjective tool, $\mathcal{G}$ is countable. We next claim that $\mathcal{G}$ is a basis of open sets. Thus fix $a \in X$ and $D \subseteq X$ open such that $a \in D$. We want to find $n, m \in \mathbb{N}$ such that $a \in B(x_n; \frac{1}{m}) \subseteq D$. Because D is open and $a \in D$, there exists $r > 0$ such that $B(a; r) \subseteq D$. By the Archimedean property, pick $k \in \mathbb{N}$ such that $\frac{1}{k} < \frac{r}{2}$. Since S is dense, it meets all the open balls in X , so $S \cap B(a; 1/k) \neq \emptyset$. Pick $n_0 \in \mathbb{N}$ such that $x_{n_0} \in B(a; 1/k)$.

We claim $a \in B(x_{n_0}; 1/k) \subseteq B(a; r) \subseteq D$. We have $x_{n_0} \in B(a; 1/k)$ implying $d(x_{n_0}, a) < 1/k$, so $d(a, x_{n_0}) < 1/k$ so $a \in B(x_{n_0}; 1/k)$. Next, let $y \in B(x_{n_0}, 1/k)$, we want to verify $y \in B(a; r)$. Therefore, $d(y, x_{n_0}) < 1/k$, so $d(a, y) \leq d(a, x_{n_0}) + d(x_{n_0}, y) < \frac{1}{k} + \frac{1}{k} = \frac{2}{k} < r$ by choice of k . Therefore, we see $d(y, a) < r$, so $y \in B(a; r)$, so $B(x_{n_0}; 1/k) \subseteq B(a; r)$. This completes the proof of the claim $a \in B(x_{n_0}; 1/k) \subseteq B(a; r) \subseteq D$, so since $B(x_{n_0}, 1/k) \in \mathcal{G}$, we have proven $\mathcal{G}$ is a basis. $\blacksquare$

Theorem 10.3:

Let (X, d) and (Y, d') be metric spaces, with $f : X \rightarrow Y$ a function. Then the following are equivalent:

- (1) f is continuous in the sense of the definition given previously
- (2) For every open set $D \subseteq Y$, $f^{-1}(D)$ is open in X .
- (3) For every closed set $F \subseteq Y$, $f^{-1}(F)$ is closed in X .

Proof:

We prove (1) $\Rightarrow$ (3) $\Rightarrow$ (2) $\Rightarrow$ (1). Suppose (1). Since f is continuous, it respects convergent sequences. Denote $f^{-1}(F) = C \subseteq X$. Fix a convergent sequence $(x_n)_{n=1}^\infty$ in C converging to a limit $b \in X$. We want to prove $b \in C$. For every $n \in \mathbb{N}$, denote $f(x_n) =: y_n \in Y$ and $f(b) =: z \in Y$. Observe that by definition of the preimage, we have $(y_n)_{n=1}^\infty$ in F . Then, observe that $x_n \rightarrow b$ implies $y_n \rightarrow z$ by continuity of f , and F is closed so it contains its limit points, so $z \in F$, therefore, $f(b) = z \in F$ implies $b \in f^{-1}(F) = C$, hence $b \in C$, so since b was an arbitrary limit point of C , we have that C is closed, completing this direction.

(3) $\Rightarrow$ (2) follows by duality of open sets and closed sets and the fact that preimages are extremely well-behaved under set theoretic operations (see lecture shell).

The idea of (2) $\Rightarrow$ (1) is to use preimages of open sets in order to verify that f satisfies the $\varepsilon - \delta$ definition of continuity (see lecture shell). $\blacksquare$

Remark 10.4:

In Theorem 10.3, it is essential to use preimages (images will not work), counterexamples exist using

$X = \mathbb{R}$ and $d = d_{usual}$. Can you give a function $g : \mathbb{R} \rightarrow \mathbb{R}$, continuous, such that $g([0, \infty))$ to $(0, 1]$, where $[0, \infty)$ is closed and $(0, 1]$ is not closed. Such an example is $g(x) = e^{-x}$.

Definition:

Given a metric space (X, d) and $\emptyset \neq A \subseteq X$, define $d_A : X \rightarrow \mathbb{R}$ by $d_A(x) := \inf\{d(x, a) : a \in A\}$. We proved on the homework that this is a contraction, i.e. $|d_A(x_1) - d_A(x_2)| \leq d(x_1, x_2)$, so in particular d_A is continuous. For $x \in X$, we have $d_A(x) = 0 \iff \inf\{d(x, a) : a \in A\} = 0 \iff \exists (a_n)_{n=1}^\infty \in A$ s.t. $a_n \rightarrow x \iff x \in \text{cl}(A)$.

Theorem 10.5 (Urysohn's Lemma):

Let (X, d) be a metric space and $A, B \subseteq X$ be non-empty closed sets such that $A \cap B = \emptyset$. Then there exists $f : X \rightarrow \mathbb{R}$ continuous such that $0 \leq f(x) \leq 1$ for all $x \in X$, $A = \{x \in X : f(x) = 0\}$, and $B = \{x \in X : f(x) = 1\}$.

Proof:

Observe that $d_A(x) + d_B(x) > 0$ for all $x \in X$, indeed, we have $d_A(x), d_B(x) \geq 0$ and $d_A(x) + d_B(x) = 0 \implies d_A(x) = 0 = d_B(x) \implies x \in \text{cl}(A) \wedge x \in \text{cl}(B) \implies x \in A \cap B = \emptyset$, which is not possible.

Define $f : X \rightarrow \mathbb{R}$ by $f(x) = \frac{d_A(x)}{d_A(x) + d_B(x)}$, where this quotient is well-defined by our above argument. We have $0 \leq f(x) \leq 1$, since the numerator is at most the denominator, and both are nonnegative. We additionally have $f(x) = 0 \iff d_A(x) = 0 \iff x \in \text{cl}(A) \iff x \in A$. Finally, $f(x) = 1 \iff d_A(x) = d_A(x) + d_B(x) \iff d_B(x) = 0 \iff x \in \text{cl}(B) \iff x \in B$. Finally, f is continuous because it respects convergent sequences: Let $(x_n)_{n=1}^\infty$ converge to $x \in X$, then $d_A(x_n) \rightarrow d_A(x)$ and $d_B(x_n) \rightarrow d_B(x)$ because they are continuous, so by limit rules for real numbers, we have $f(x_n) = \frac{d_A(x_n)}{d_A(x_n) + d_B(x_n)} \rightarrow \frac{d_A(x)}{d_A(x) + d_B(x)} = f(x)$. ■

Definition 11.1:

Let (X, d) be a metric space, and let $(x_n)_{n=1}^\infty$ be a sequence of elements in X . We say that $(x_n)_{n=1}^\infty$ is a Cauchy sequence, when, for all $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that for $n, m \geq n_0$, we have $d(x_n, x_m) < \varepsilon$.

Proposition 11.3:

Let (X, d) be a metric space, and let $(x_n)_{n=1}^\infty$ be a convergent sequence in X . Then $(x_n)_{n=1}^\infty$ is Cauchy.

Proof:

Let $\varepsilon > 0$. We know there exists $a \in X$ such that $\lim_{n \rightarrow \infty} x_n = a$. Therefore, by definition of convergence, there exists $n_0 \in \mathbb{N}$ such that for $n \geq n_0$, we have $d(x_n, a) < \varepsilon/2$, so by the triangle inequality, for $n, m \geq n_0$, we have $d(x_m, x_n) \leq d(x_m, a) + d(a, x_n) < \varepsilon/2 + \varepsilon/2 = \varepsilon$. ■

Definition 11.4:

When the converse to Proposition 11.3 holds, that is, every Cauchy sequence is convergent in (X, d) , then we say that (X, d) is complete.

Definition:

A Polish space is a metric space (X, d) that is separable and complete.

Definition 11.5:

Let $(X, \|\cdot\|)$ be a normed vector space. If d is the distance induced by $\|\cdot\|$, and (X, d) is complete, we say that $(X, \|\cdot\|)$ is a Banach space.

Example:

For any $k \in \mathbb{N}$, $1 \leq p \leq \infty$, $(\mathbb{R}^k, \|\cdot\|_p)$ is a Banach space.

Example 11.6:

$(\mathbb{R}_{\text{fin}}^\infty, \rho)$, where ρ is the distance induced by $\|\cdot\|_\infty$, is not complete! How do we find a sequence $(x_n)_{n=1}^\infty$ which is Cauchy but not convergent? Let $x_1 = (1, 0, 0, \dots)$, $x_2 = (1, 1/2, 0, 0, \dots)$, $x_n =$

$(1, 1/2, 1/3, \dots, 1/n, 0, 0, \dots)$. This sequence is Cauchy, (quiz), but is not convergent: as proved on an assignment, convergence under ρ implies componentwise convergence, i.e. there exists $a \in \mathbb{R}_{\text{fin}}^\infty$ such that $a^{(k)} = \lim_{n \rightarrow \infty} x_n^{(k)}$ for every $k \in \mathbb{N}$. But $x_n^{(k)} = 1/k$ for all $n \geq k$, which implies $a^{(k)} = 1/k$ for all $k \in \mathbb{N}$, so every component of a is nonzero, so $a \notin \mathbb{R}_{\text{fin}}^\infty$, contradiction, so we cannot have convergence.

As a natural question, what if we add “new” vectors to $\mathbb{R}_{\text{fin}}^\infty$ to make it complete? This leads toward the idea of the completion of a metric space. We do however, have a partial converse:

Proposition 11.7:

Let (X, d) be a metric space, and let $(x_n)_{n=1}^\infty$ be a sequence in X . Suppose that $(x_n)_{n=1}^\infty$ is Cauchy and has a cluster point $a \in X$. Then $(x_n)_{n=1}^\infty \rightarrow a$.

Proof:

Let $\varepsilon > 0$. Because $(x_n)_{n=1}^\infty$ is Cauchy, there exists $n_1 \in \mathbb{N}$ such that for $n, m \geq n_1$, we have $d(x_m, x_n) < \varepsilon/2$. Because a is a cluster point, $(x_n)_{n=1}^\infty$ visits the ball $B(a; \varepsilon/2)$ infinitely many times, hence there exists $n_0 \geq n_1$ such that $x_{n_0} \in B(a; \varepsilon/2)$, that is, $d(x_{n_0}, a) < \varepsilon/2$. But now for $n \geq n_0$, we have $d(x_n, a) \leq d(x_n, x_{n_0}) + d(x_{n_0}, a) < \varepsilon/2 + \varepsilon/2 = \varepsilon$. ■

Note that if $(x_n)_{n=1}^\infty$ is Cauchy, then $d(x_n, x_{n+1}) \rightarrow 0$, but the converse is not true, take x_n to be the n th partial sum of the harmonic series under the metric space $(\mathbb{R}, d_{\text{std}})$, where see $d_{\text{std}}(x_n, x_{n+1}) \rightarrow 0$ but this sequence is divergent and thus not Cauchy (using completeness of $\mathbb{R}$).

Exercise 11.9:

A useful fact is that if for some $\alpha \in (0, \infty)$, $\gamma \in (0, 1)$, and $d(x_n, x_{n+1}) \leq \alpha\gamma^n$ for all $n \in \mathbb{N}$, then $(x_n)_{n=1}^\infty$ is Cauchy. The relevant point is that $\gamma < 1$, so $\sum_{n=1}^\infty \gamma^n = \frac{\gamma}{1-\gamma} < \infty$.

As a special case,

Definition 11.10:

A sequence $(x_n)_{n=1}^\infty$ is 1/2-geometric when $d(x_n, x_{n+1}) \leq \frac{1}{2^n}$ for all $n \in \mathbb{N}$, which by the above exercise, must be Cauchy.

Our idea is to only check if the 1/2-geometric sequences are convergent to tell us whether our metric space is complete or not:

Lemma 11.11:

Let (X, d) be a metric space, and let $(x_n)_{n=1}^\infty$ be a Cauchy sequence in X . Then it has a subsequence $(x_{n(k)})_{k=1}^\infty$ which is 1/2-geometric.

Proof:

We start by invoking the Cauchy condition for $\varepsilon = 1/2$, in order to find an $n(1) \in \mathbb{N}$ such that $m, n \geq n(1) \implies d(x_m, x_n) < 1/2$. Recursively, for all $k \in \mathbb{N}$ pick $n(k) \in \mathbb{N}$ such that $m, n \geq n(k) \implies d(x_m, x_n) < 1/2^{k+1}$ and $n(k) > n(k-1)$. The subsequence of $(x_n)_{n=1}^\infty$ given by picking these indices works. ■

Proposition 11.12:

Let (X, d) be a metric space. If every 1/2-geometric sequence in X has a cluster point, then (X, d) is complete.

Proof:

Let $(x_n)_{n=1}^\infty$ be a Cauchy sequence in X . By the above lemma, it has a subsequence $(x_{n(k)})_{k=1}^\infty$ that is 1/2-geometric. But the cluster points of this subsequence are also cluster points of the overall sequence, so our Cauchy sequence must have a cluster point, so by Proposition 11.7, it converges to that point, and so we have proven our arbitrary Cauchy sequence converges. ■

Lemma 11.14:

Let (X, d) be a complete metric space, and let $f : X \rightarrow X$ be a strict contraction. Let $x_1 \in X$. Then,

defining the recursive sequence $x_k = f(x_{k-1})$ for all $k \geq 2 \in \mathbb{N}$, the sequence $(x_k)_{k=1}^\infty$ converges.

Proof:

Homework 7A. ■

Theorem 11.13 (Banach Contraction Principle):

Let (X, d) be a complete metric space, and let $f : X \rightarrow X$ be a strict contraction. Then there exists a unique $p \in X$ such that $f(p) = p$.

Proof:

Let $x_1 \in X$, and recursively define $x_k = f(x_{k-1})$ for all $k \geq 2 \in \mathbb{N}$. Invoke the above lemma to get a limit $\lim_{n \rightarrow \infty} x_n =: p \in X$. We claim $f(p) = p$. Indeed, let $n \in \mathbb{N}$. Then, we have

$$d(p, f(p)) \leq d(p, x_n) + d(x_n, f(x_n)) + d(f(x_n), f(p)) = d(p, x_n) + d(x_n, x_{n+1}) + d(f(x_n), f(p))$$

c is Lip. constant, rest is def. of f

$$\underbrace{\leq}_{\leq} d(x_n, p) + (d(x_n, p) + d(p, x_{n+1})) + cd(x_n, p) = (2+c)d(x_n, p) + d(x_{n+1}, p) \rightarrow 0$$

giving that $d(p, f(p)) = 0$ by the squeeze theorem, so $p = f(p)$, proving existence.

For uniqueness, suppose there exist $p, q \in X$ such that $p = f(p)$ and $q = f(q)$. Then,

$$d(p, q) = d(f(p), f(q)) \leq cd(p, q) \implies (1-c)d(p, q) \leq 0 \implies d(p, q) \leq 0 \implies d(p, q) = 0$$

giving $p = q$. ■

Definition 12.1

We say that a metric space (X, d) has the nested closed balls property if whenever $x_1, x_2, \dots, x_n, \dots \in X$ and $r_1, r_2, \dots, r_n, \dots > 0$ are such that $\lim_{n \rightarrow \infty} r_n = 0$ and such that $\overline{B}(x_1; r_1) \supseteq \overline{B}(x_2; r_2) \supseteq \dots \supseteq \overline{B}(x_n; r_n) \supseteq \dots$, we have $\bigcap_{n=1}^\infty \overline{B}(x_n; r_n) \neq \emptyset$.

Theorem 12.3:

(X, d) is complete in the sense of Definition 11.4 if and only if it has the nested closed balls property in the sense of Definition 12.1.

Proof:

In the forward direction, suppose we have $x_1, x_2, \dots, x_n, \dots \in X$ and $r_1, r_2, \dots, r_n, \dots \in (0, \infty)$ with $r_n \rightarrow 0$ such that $\overline{B}(x_1; r_1) \supseteq \overline{B}(x_2; r_2) \supseteq \dots$. We claim the sequence $(x_n)_{n=1}^\infty$ is Cauchy. Fix $\varepsilon > 0$. Since $r_n \rightarrow 0$, there exists $n_0 \in \mathbb{N}$ such that for $n \geq n_0$, we have $r_n < \varepsilon/2$. Now, for $n, m \geq n_0$, $x_n \in \overline{B}(x_n; r_n) \subseteq \overline{B}(x_{n_0}; r_{n_0})$ implying $d(x_n, x_{n_0}) \leq r_{n_0} < \varepsilon/2$, and likewise $d(x_m, x_{n_0}) < \varepsilon/2$, so by the triangle inequality $d(x_n, x_m) \leq d(x_n, x_{n_0}) + d(x_m, x_{n_0}) < \varepsilon$. Therefore, $(x_n)_{n=1}^\infty$ is Cauchy, and thus convergent since (X, d) is complete. Take $a = \lim_{n \rightarrow \infty} x_n$. We claim $a \in \bigcap_{n=1}^\infty \overline{B}(x_n; r_n)$. Indeed, fix $n \in \mathbb{N}$. Then, for every $m \geq n$, $d(a, x_n) \leq d(a, x_m) + d(x_m, x_n)$. We have $d(x_m, x_n) \leq r_n$, since $x_m \in \overline{B}(x_m; r_m) \subseteq \overline{B}(x_n; r_n)$ by the nested property of our balls, so our estimate becomes $d(a, x_n) \leq d(a, x_m) + r_n$. Then, taking $m \rightarrow \infty$, $d(a, x_m) \rightarrow 0$ by convergence, giving $d(a, x_n) \leq r_n$. Therefore, $a \in \overline{B}(x_n; r_n)$, so since n was arbitrary, this gives $a \in \bigcap_{n=1}^\infty \overline{B}(x_n; r_n)$, completing this direction.

Conversely, we will verify all $1/2$ -geometric sequences in X converge. Thus, pick $(x_n)_{n=1}^\infty$ in X such that $d(x_n, x_{n+1}) < 1/2^n$ for all $n \in \mathbb{N}$. First, for every $n \in \mathbb{N}$, we have $\overline{B}(x_{n+1}; 1/2^n) \subseteq \overline{B}(x_n; 1/2^{n-1})$ by the triangle inequality. By the nested closed balls property (nested by our above inclusion, and the remaining hypothesis holds since $1/2^n \rightarrow 0$), this means we have $a \in \bigcap_{n=1}^\infty \overline{B}(x_n, 1/2^{n-1})$. This a works as the limit of the sequence $(x_n)_{n=1}^\infty$, since for every $n \in \mathbb{N}$, $0 \leq d(a, x_n) \leq \frac{1}{2^{n-1}}$, which by the squeeze theorem tells us $\lim_{n \rightarrow \infty} d(a, x_n) = 0$ so $x_n \rightarrow a$. ■

Proposition 12.4:

Let (X, d) be a complete metric space and let Y be a (nonempty) closed subset of X . Then, denoting the restriction of d to Y as d_o , we have that (Y, d_o) is complete.

Proof:

Consider a Cauchy sequence in Y . By completeness of X , this converges to a limit in X , which must be

in Y by closure. Therefore every Cauchy sequence in Y converges to a limit in Y . ■

Now, we will use the space $F_b(M, \mathbb{R})$ as the bigger complete space given. This is indeed complete (see lecture shell for sketch of proof, using the nested-closed-balls property). This is a truly “big space”, it is not separable for instance.

Missed lecture, intro to G_δ and F_σ sets.

It is easy to see that $\mathbb{Q}$ is of type F_σ , but is it of type G_δ ? At the moment, it's hard to see whether any sets are not of both categories. Baire's theorem will answer this question.

Remark 13.4:

Let (X, d) be a metric space. Suppose $D_1, D_2 \subseteq X$ are open and dense in X , i.e. $\text{cl}(D_1) = \text{cl}(D_2)$. Then $D := D_1 \cap D_2$ is clearly open, and moreover is dense in X . How do we check that D is dense in X ? D being dense in X can be described as $D \cap B(a; r) \neq \emptyset$ for all $a \in X$ and $r \in (0, \infty)$. So fix $a \in X$ and $r \in (0, \infty)$. Since D_1 is dense in X , $D_1 \cap B(a; r) \neq \emptyset$, so pick $x_1 \in D_1 \cap B(a; r)$ (hi axiom of choice). Observe $D_1 \cap B(a; r)$ is open, and so since $x_1 \in D_1 \cap B(a; r)$, there exists $r_1 > 0$ such that $B(x_1; r_1) \subseteq D_1 \cap B(a; r)$. Now, by density of D_2 , we have $D_2 \cap B(x_1; r_1) \neq \emptyset$, so pick a point $x_2 \in D_2 \cap B(x_1; r_1)$. Then, since $D_2 \cap B(x_1; r_1) \subseteq D_2 \cap (D_1 \cap B(a; r)) = (D_1 \cap D_2) \cap B(a; r) = D \cap B(a; r)$, so $x_2 \in D \cap B(a; r)$, so the intersection is not empty, showing D is dense. ■

From this, an immediate induction gives if $D_1, \dots, D_n \subseteq X$ are open and dense in X , then $D_1 \cap \dots \cap D_n$ is open and dense in X . This density generalizes as we take countably many sets:

Theorem 13.5 (Baire):

Suppose (X, d) is a complete metric space. Let $D_1, \dots, D_n, \dots \subseteq X$ be open and dense in X . Then $\bigcap_{n=1}^{\infty} D_n$ is dense in X .

Note: $\bigcap_{n=1}^{\infty} D_n$ may no longer be open, but it is of type G_δ . In short: $\bigcap_{n=1}^{\infty} D_n$ is a “dense G_δ ”.

Proof:

As a motivating idea, we want to do the same proof as above where we take successive smaller open balls $B(x_1; r_1), B(x_2; r_2), \dots, B(x_n, r_n), \dots$ and use completeness, but this isn't guaranteed to work. We need to use a trick to let us handle nested open balls (which is lemma 13.10, question 2 on homework 7B), and we will be done with the same proof as above. TODO fill in more from lecture shell. ■

Corollary 13.6:

Let (X, d) be a complete metric space. Suppose $A_1, \dots, A_n \subseteq X$ are all dense and of type G_δ . Then $\bigcap_{k=1}^n A_k$ is a dense G_δ .

Proof:

Since $A_1, \dots, A_n$ are G_δ 's, we can write $A_1 = D_1^{(1)} \cap D_2^{(1)} \cap \dots \cap D_k^{(1)} \cap \dots$, and in general $A_k = D_1^{(k)} \cap D_2^{(k)} \cap \dots \cap D_k^{(k)} \cap \dots$. Thus, since for every $1 \leq m \leq n$ and $k \in \mathbb{N}$ we have $A_m \subseteq D_k^{(m)}$, so $\text{cl}(A_m) \subseteq \text{cl}(D_k^{(m)})$, but $\text{cl}(A_m) = X$, so we see $D_k^{(m)}$ is dense. Now write $A_1 \cap \dots \cap A_n = D_1^{(1)} \cap \dots \cap D_1^{(n)} \cap D_2^{(1)} \cap \dots \cap D_2^{(n)} \cap \dots \cap D_k^{(1)} \cap \dots \cap D_k^{(n)} \cap \dots$, i.e. $A_1 \cap \dots \cap A_n$ is a countable union of dense open sets, so $A_1 \cap \dots \cap A_n$ is dense by Baire's theorem. ■

The above proof can be adjusted to give:

Corollary 13.7:

Let (X, d) be a complete metric space. Suppose $A_1, \dots, A_n, \dots \subseteq X$ where every A_n is a dense G_δ , then $\bigcap_{n=1}^{\infty} A_n$ is a dense G_δ .

Example 13.8:

Take $X = \mathbb{R}$, $d = d_{\text{usual}}$, and take $Q \subseteq \mathbb{R}$. By Corollary 13.6 (with the sets $A_1 = \mathbb{Q}, A_2 = \mathbb{R} \setminus \mathbb{Q}$) and other observations, we can prove $\mathbb{Q}$ is not a set of type G_δ .

The lemma we use to prove Baire's theorem is as follows:

Lemma 13.10:

Let (X, d) be a complete metric space, Let $(x_n)_{n=1}^\infty$ be a sequence of elements of X and let $(r_n)_{n=1}^\infty$ be a sequence in $(0, \infty)$ such that $\lim_{n \rightarrow \infty} r_n = 0$, and

$$B(x_2; r_2) \subseteq B(x_1; \frac{r_1}{2}), B(x_3; r_3) \subseteq B(x_2; \frac{r_2}{2}), \dots$$

Then $\bigcap_{n=1}^\infty B(x_n; r_n) \neq \emptyset$.

Proof:

Question 2 on Assignment 7B. ■

Theorem 14.1 (Baire):

Let (X, d) be a complete metric space. Let $F_1, F_2, \dots, F_n, \dots \subseteq X$ be closed. Suppose $\text{Int}(\bigcup_{n=1}^\infty F_n) \neq \emptyset$. Then there exists $n_0 \in \mathbb{N}$ such that $\text{Int}(F_{n_0}) \neq \emptyset$.

Remark:

We always have $\text{Int}(\bigcup_{n=1}^\infty F_n) \supseteq \bigcup_{n=1}^\infty \text{Int}(F_n)$, as an example, take $X = \mathbb{R}$, d to be the usual distance, and take $F_1 = [0, 1]$, $F_2 = [1, 2]$, generally $F_n = [n-1, n]$. Then $\text{Int}(\bigcup_{n=1}^\infty F_n) = \text{Int}([0, \infty)) = (0, \infty)$, and $\bigcup_{n=1}^\infty \text{Int}(F_n) = \bigcup_{n=1}^\infty (n-1, n) = (0, \infty) \setminus \mathbb{N}$.

Proof:

Assume contrapositively that $\text{Int}(F_n) = \emptyset$ for all $n \in \mathbb{N}$. For every $n \in \mathbb{N}$, put $D_n = X \setminus F_n$, which is open since F_n is closed. Observe that $\text{cl}(D_n) = \text{cl}(X \setminus F_n) = X \setminus \text{Int}(F_n) = X \setminus \emptyset = X$ by duality, so D_n is dense in X for every $n \in \mathbb{N}$. Thus, by Baire's Theorem above, $\bigcap_{n=1}^\infty D_n$ is dense in X . Hence $\text{cl}(\bigcap_{n=1}^\infty D_n) = X$, but then compute using DeMorgan's laws and duality, $\text{Int}(\bigcup_{n=1}^\infty F_n) = \text{Int}(\bigcup_{n=1}^\infty (X \setminus D_n)) = \text{Int}(X \setminus \bigcap_{n=1}^\infty D_n) = X \setminus \text{cl}(\bigcap_{n=1}^\infty D_n) = X \setminus X = \emptyset$, which is the negation of our hypothesis. ■

Exercise 14.6:

Take $X = \mathbb{R}$, d to be the usual distance. Suppose $f_1, f_2, \dots, f_n, \dots : \mathbb{R} \rightarrow \mathbb{R}$ are continuous functions. Suppose that for all $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $f_n(x) \in \mathbb{Q}$. Then there exists an open interval $(a, b) \subseteq \mathbb{R}$ and $n \in \mathbb{N}$ such that f_n is constant on (a, b) .

Proof:

Enumerate $\mathbb{Q} = \{q_1, q_2, \dots, q_n, \dots\}$. For every $m, n \in \mathbb{N}$, denote $F_{m,n} = \{x \in \mathbb{R} \mid f_n(x) = q_m\}$. Our hypothesis gives $\bigcup_{m,n \in \mathbb{N}} F_{m,n} = \mathbb{R}$. Observe every $F_{m,n}$ is a closed set, indeed, we can write $F_{m,n} = f_n^{-1}(\{q_m\})$, hence $F_{m,n}$ is the preimage of the closed set $\{q_m\}$ under a continuous function f_n , so is closed. So we have a countable family $(F_{m,n})_{(m,n) \in \mathbb{N} \times \mathbb{N}}$ of closed subsets such that $\text{Int}(\bigcup_{m,n \in \mathbb{N}} F_{m,n}) = \text{Int}(\mathbb{R}) = \mathbb{R} \neq \emptyset$, so by Theorem 14.1, there must exist $m_o, n_o \in \mathbb{N}$ such that $\text{Int}(F_{m_o, n_o}) \neq \emptyset$. For this $m_o, n_o \in \mathbb{N}$ let us pick a point $c \in \text{Int}(F_{m_o, n_o})$ and $r > 0$ such that $(c-r, c+r) \subseteq F_{m_o, n_o} = \{x \mid f_{n_o}(x) = q_{m_o}\}$. ■

Exercise 14.7:

Let $(X, \|\cdot\|)$ be a Banach space (completeness is important here). Let $T : X \rightarrow X$ be linear and continuous. Define $T_1 = T$, $T_2 = T \circ T$, more generally $T_n = \underbrace{T \circ T \circ \dots \circ T}_{n \text{ times}}$. Then for all $n \in \mathbb{N}$, $T_n : X \rightarrow X$

is linear and continuous. Suppose that for every $x \in X$, there exists $n \in \mathbb{N}$ such that $T_n(x) = x$. Prove there exists $n_0 \in \mathbb{N}$ such that $T_{n_0}(x) = x$ for all $x \in X$.

Proof:

For every $n \in \mathbb{N}$, define $V_n = \{x \in X \mid T_n(x) = x\}$, which is a linear subspace of X , and is also closed (since the set of fixed points of a linear continuous function on a Banach space is always a subspace). We have as an equivalent formulation of our hypothesis, that $\bigcup_{n=1}^\infty V_n = X$. So by Theorem 14.1, since $\text{Int}(\bigcup_{n=1}^\infty V_n) = \text{Int} X = X \neq \emptyset$, so there must exist $n_0 \in \mathbb{N}$ such that $\text{Int}(V_{n_0}) \neq \emptyset$. By Question 5 on Assignment 4 (Problem P6.3), this implies $V_{n_0} = X$ since V_{n_0} is a linear subspace of X with nonempty interior. Thus, by definition of V_{n_0} , for every $x \in V_{n_0} = X$, we have $T_{n_0}(x) = x$. ■

Example:

Consider the metric space $(\mathbb{R}_{\text{fin}}^\infty, \|\cdot\|_\infty)$. Define $T : \mathbb{R}_{\text{fin}}^\infty \rightarrow \mathbb{R}_{\text{fin}}^\infty$ as follows: for $x = (x^{(1)}, x^{(2)}, \dots)$, define $T(x) = y = (x^{(1)}, x^{(3)}, x^{(2)}, x^{(6)}, x^{(5)}, x^{(4)}, \dots)$, more generally we keep on doing a bigger and bigger cycle of x 's components to get y 's components. Then, T is linear and continuous, as can be straightforwardly checked. Moreover, every $x \in \mathbb{R}_{\text{fin}}^\infty$ is periodic for T . Say for example $x^{(k)} = 0$ for $k > 10$. Then $T_{12}(x) = x$ (using the notation of the above exercise). But there is no $n_0 \in \mathbb{N}$ such that $T_{n_0}(x) = x$ for all $x \in \mathbb{R}_{\text{fin}}^\infty$, since we can take arbitrarily long $x \in \mathbb{R}_{\text{fin}}^\infty$ to force n_0 bigger and bigger (since n_0 must be a multiple of the cycle lengths that make up x 's length of nonzero components). This shows how important the completeness hypothesis to Exercise 14.7 is (we invoked Baire's theorem above).

Definition 15.1:

Let (X, d) be a metric space, and let $A \subseteq X$. An open cover of A is a family $(D_i)_{i \in I}$ (with no restrictions on the index set I) of open subsets of X , such that $A \subseteq \bigcup_{i \in I} D_i$.

Definition 15.2:

Let (X, d) be a metric space, and let A be a subset of X . We say that A is compact if for every open cover $(D_i)_{i \in I}$, A has a finite subcover, that is, there exists $i_1, \dots, i_n \in I$ such that $A \subseteq D_{i_1} \cup \dots \cup D_{i_n}$.

Remarks/Examples 15.3:

- (1) This definition depends only on the topology of (X, d) (we only use open sets), and is used in the more general framework of topological spaces.
- (2) Every finite set is compact.
- (3) In calculus, we had compact if and only if closed and bounded (Heine-Borel). It is still true that compactness implies the set is closed and bounded, but the converse is not true, we need something stronger: replace closed with complete, and bounded with a property we will call totally bounded.
- (4) Compact sets are “not too big”, for instance we will prove compactness implies separability.

Definition 15.4/Remark 15.5:

Let (X, d) be a metric space, and let A be a non-empty subset of X . We say that A is totally bounded if for all $\varepsilon > 0$, there exist $a_1, \dots, a_m \in A$ such that $B(a_1; \varepsilon) \cup \dots \cup B(a_m; \varepsilon) \supseteq A$. (We make the convention the empty set is totally bounded). Such a list of points $a_1, \dots, a_m$ is called a (finite) ε -net of A . It is sometimes convenient to state their condition equivalently as $\min\{d(a, a_1), \dots, d(a, a_m)\} < \varepsilon$ for all $a \in A$. With this terminology, a set A is totally bounded if and only if it has a finite ε -net for all $\varepsilon > 0$.

Remark:

Let (X, d) be a metric space, with $\emptyset \neq A \subseteq X$ a totally bounded set. For every $\varepsilon > 0$, it makes sense to define

$$N_\varepsilon(A) = \min\{p \in \mathbb{N} \mid \text{an } \varepsilon\text{-net } a_1, \dots, a_p \text{ of } A \text{ exists}\}$$

We can look in particular at $N_{1/k}(A)$ for $k \in \mathbb{N}$. Suppose we have found some $c, d \in (0, \infty)$ such that $N_{1/k}(A) \sim ck^d$ as $k \rightarrow \infty$. The informal idea is that the d that has appeared could maybe serve as a definition of the dimension of the set (this is called the “box dimension” of a totally bounded set). Thus, suppose $\lim_{k \rightarrow \infty} \frac{N_{1/k}(A)}{ck^d} = 1$. How do we try to calculate d ? Taking logarithms, we have $\log N_{1/k}(A) - \log ck^d = \log\left(\frac{N_{1/k}(A)}{ck^d}\right) \rightarrow \log 1 = 0$. So we get $\log N_{1/k}(A) - \log c - d \log k \rightarrow 0$, which

dividing by $\log k$, tells us $\log N_{1/k}(A) / \log k - \overbrace{\log c / \log k}^{\rightarrow 0} - d \rightarrow 0$ hence $\frac{\log N_{1/k}(A)}{\log k} \rightarrow d$.

Definition:

For any A as above, if the limit $\frac{\log N_{1/k}(A)}{\log k}$ exists, then we call it the box-dimension of A .

Proposition 15.6:

Let (X, d) be a metric space with $A \subseteq X$. If A is compact, then A is totally bounded.

Proof:

Let $\varepsilon > 0$. We want to find $a_1, \dots, a_p \in A$ such that $B(a_1; \varepsilon) \cup \dots \cup B(a_p; \varepsilon) \supseteq A$. Denote $D_a := B(a; \varepsilon)$ for every $a \in A$. Then $(D_a)_{a \in A}$ is an open cover of A . By compactness, we can find $a_1, \dots, a_p \in A$ such that $D_{a_1}, \dots, D_{a_p}$ cover A , hence $a_1, \dots, a_p$ is a finite ε -net. ■

Remark/Proposition 15.7:

Let (X, d) be a metric space, and $A \subseteq X$ a totally bounded set. Then A is separable and bounded. The latter of these is a homework problem. For the former of these:

For every $k \in \mathbb{N}$, pick a finite $\frac{1}{k}$ -net in A , call it $a_1^{(k)}, \dots, a_{p_k}^{(k)}$. Let $S = \{a \in A \mid \exists k \in \mathbb{N} \text{ and } 1 \leq i \leq p_k \text{ such that } a = a_i^{(k)}\}$.

Then $S \subseteq A$ is countable. We claim $\text{cl}(S) \supseteq A$: Pick a point $a \in A$, and let $r > 0$, for which we verify $B(a; r) \cap S \neq \emptyset$. By the Archimedean property, choose $k \in \mathbb{N}$ such that $\frac{1}{k} < r$, and observe $B(a; \frac{1}{k}) \cap \{a_1^{(k)}, \dots, a_{p_k}^{(k)}\} \neq \emptyset$ by construction of the net, so $B(a; \frac{1}{k}) \cap S \neq \emptyset$, implying $B(a; r) \cap S \neq \emptyset$, so a is adherent, so $a \in \text{cl}(S)$. ■

Definition 15.9:

Let (X, d) be a metric space, and let A be a nonempty subset of X . We say A is complete when every Cauchy sequence $(x_n)_{n=1}^\infty$ in A is convergent to a limit in A (and make the convention that the empty set is complete).

Remark 15.10:

Observe that every complete set is closed (as it contains its limit points). Conversely,

Proposition 15.11:

If (X, d) is a complete metric space, then all closed sets are complete.

Proof:

If $(x_n)_{n=1}^\infty$ is a Cauchy sequence in A , then it is a Cauchy sequence in X , and (X, d) is complete so the sequence converges. But A is closed, so it contains the limits of all of its convergent sequences. Therefore, we see every Cauchy sequence in A converges (by completeness of X) to a limit in A (by closure). ■

Proposition 15.12:

Let (X, d) be a metric space with $A \subseteq X$. Then A being compact implies A is complete.

Proof:

Let $(x_n)_{n=1}^\infty$ be a Cauchy sequence in A . By Proposition 11.7, it will suffice to verify that there exists $a \in A$ such that a is a cluster point for $(x_n)_{n=1}^\infty$. Assume by contradiction that no $a \in A$ is a cluster point for $(x_n)_{n=1}^\infty$. Therefore, for all $a \in A$, we can find $r(a) > 0$ and $n(a) \in \mathbb{N}$ such that $x_n \notin B(a; r(a))$ for $n > n(a)$. But now, $(B(a; r(a)))_{a \in A}$ is an open cover for A , so since A is compact, we can select a finite number of points $a_1, \dots, a_p \in A$ such that $B(a_1; r(a_1)) \cup \dots \cup B(a_p; r(a_p)) \supseteq A$. Define $n_0 = 1 + \max\{n(a_1), \dots, n(a_p)\}$ and look at x_{n_0} . We have $x_{n_0} \in A$, but $n_0 > n(a_j)$ for $j \in \{1, \dots, p\}$ implies $x_{n_0} \notin B(a_j; r(a_j))$, hence $x_{n_0} \notin B(a_1; r(a_1)) \cup \dots \cup B(a_p; r(a_p)) \supseteq A$, a contradiction. ■

Therefore, we see when A is compact, it is totally bounded and complete. The converse is also true, which we will work our way up to proving.

Exercise:

Consider the metric space $(\mathbb{R}_0^\infty, \rho)$. Let $C = \{x \in \mathbb{R}_0^\infty : |x^{(j)}| \leq \frac{1}{j} \ \forall j \in \mathbb{N}\}$. Let $\varepsilon = \frac{1}{10}$. How many vectors in C do we need in order to form a finite ε -net (and explain how you find such a net, if we were to generalize our choice of ε).

Definition 16.1:

Let (X, d) be a metric space. We say A is sequentially compact if every sequence $(x_n)_{n=1}^\infty$ in A has a cluster point in A .

Proposition 16.3 (Lebesgue covering lemma):

Let (X, d) be a metric space and let $A \subseteq X$ be sequentially compact. Let $(D_i)_{i \in I}$ be an open cover of A . Then there exists $r > 0$ such that for every $a \in A$, there exists $i \in I$ such that $B(a; r) \subseteq D_i$ (such a radius is called a Lebesgue radius).

Proof:

For a given $a \in A$, say $r > 0$ “pleases” a when there exists $i_o \in I$ such that $B(a; r) \subseteq D_{i_o}$.

We will show we can find an $r = 1/n$ for some $n \in \mathbb{N}$. Assume by contradiction that this is not true. Then for every $n \in \mathbb{N}$, there exists $a_n \in A$ such that $B(a_n; 1/n) \not\subseteq D_i$ for all $i \in I$. TODO see sample homework solution. ■

Lemma 16.6 (the weeding lemma):

Let (X, d) be a metric space and let A be a totally bounded subset of X . Suppose $(x_n)_{n=1}^\infty$ is a sequence in A , we have an infinite subset $S \subseteq \mathbb{N}$, and let $\varepsilon > 0$. Then we can find a subset $S' \subseteq S$ still infinite such that $d(x_m, x_n) < \varepsilon$ for every $m, n \in S'$.

Proof:

Let $a_1, \dots, a_p$ be a finite $\varepsilon/2$ -net for A , so for every $x_n \in A$, we have $x_n \in B(a_1; \varepsilon/2) \cup \dots \cup B(a_p; \varepsilon/2)$. Thus we have

$$S = \{n \in S \mid x_n \in B(a_1; \varepsilon/2)\} \cup \dots \cup \{n \in S \mid x_n \in B(a_p; \varepsilon/2)\}$$

Since S is infinite, by the pigeonhole principle, there exists $j \in \{1, \dots, p\}$ such that $S' = \{n \in S \mid x_n \in B(a_j; \varepsilon/2)\}$ is an infinite set. This S' works, where by the triangle inequality, for any $x_m, x_n \in S'$, we have $d(x_m, x_n) \leq d(x_m; a_j) + d(x_n; a_j) < \varepsilon$. ■

Theorem 16.2:

Let (X, d) be a metric space with $A \subseteq X$. Then the following are equivalent:

- (1) A is compact
- (2) A is sequentially compact
- (3) A is totally bounded and complete.

Proof:

We have already proved (1) implies (3). Now, assume (2), we prove (1). Let $(D_i)_{i \in I}$ be an open cover of A , and assume by contradiction that it does not have a finite subcover, that is, for all $n \in \mathbb{N}$ and $i_1, \dots, i_n \in I$, we have $A \not\subseteq D_{i_1} \cup \dots \cup D_{i_n}$, that is, $A \setminus (D_{i_1} \cup \dots \cup D_{i_n}) \neq \emptyset$. By Lebesgue's covering lemma, there exists $r > 0$ that satisfies the properties given in Proposition 16.3. Now, pick any point $a_1 \in A$ and pick an $i_1 \in I$ such that $D_{i_1} \supseteq B(a_1; r)$ (since r pleases a_1 in the terminology of the proof above). But we know $A \setminus D_{i_1} \neq \emptyset$, hence pick $a_2 \in A \setminus D_{i_1}$ and pick $i_2 \in I$ such that $D_{i_2} \supseteq B(a_2; r)$. In general, assume we have $a_1, \dots, a_n \in A$ and $i_1, \dots, i_n \in I$, and pick $a_{n+1} \in A \setminus (D_{i_1} \cup \dots \cup D_{i_n})$ (since this set isn't empty), and pick $i_{n+1} \in I$ such that $D_{i_{n+1}} \supseteq B(a_{n+1}; r)$.

Observe that $d(a_{n+1}, a_m) \geq r$ for all $1 \leq m \leq n$, since $a_{n+1} \in A \setminus (D_{i_1} \cup \dots \cup D_{i_n})$ implies $a_{n+1} \notin D_{i_m}$ (with $D_{i_m} \supseteq B(a_m; r)$) implying $a_{n+1} \notin B(a_m; r)$.

Thus, we have a sequence $a_1, \dots, a_n, \dots \in A$ such that $d(a_n, a_m) \geq r$ for all $n \neq m$. This means $(a_n)_{n=1}^\infty$ is an r -repelling sequence, so such a sequence does not have any cluster points by Question 2 on Assignment 3. This contradicts sequential compactness, completing the proof of this direction.

Now, assume (3). We prove (2). Let $(x_n)_{n=1}^\infty$ be a sequence in A . We claim we can find a subsequence $(x_{n(k)})_{k=1}^\infty$ that is Cauchy. To see this, use the weeding lemma (16.6) to construct subsets $S_0 \supseteq S_1 \supseteq \dots \supseteq S_k \supseteq \dots$ of $\mathbb{N}$. Let $S_0 = \mathbb{N}$. Take $S_1 \subseteq S_0$ to be the set S' we get from Lemma 16.6 with $\varepsilon = \frac{1}{2}$, and inductively $S_{k+1} \subseteq S_k$ from Lemma 16.6 with $\varepsilon = \frac{1}{2^{k+1}}$. Now, construct indices $1 \leq n(1) < n(2) < \dots < n(k) < \dots$ as follows: $n(1)$ is an arbitrary element of S_1 , $n(2)$ is an arbitrary element of S_2 that is bigger than $n(1)$ (using infinite size of S_2), and generally $n(k)$ is an arbitrary element of S_k bigger than $n(k-1)$.

Observe by construction for every $k \in \mathbb{N}$ we have that $d(x_{n(k)}, x_{n(k+1)}) < \frac{1}{2^k}$, which tells us the subsequence $(x_{n(k)})_{k=1}^\infty$ is $\frac{1}{2}$ -geometric, and thus Cauchy by our lecture on completeness.

Now, because the sequence $(x_{n(k)})_{k=1}^\infty$ is Cauchy and A is complete, it has a convergent limit in A .

Therefore, a subsequence of $(x_n)_{n=1}^\infty$ has a limit in A , so the sequence $(x_n)_{n=1}^\infty$ has this limit as a cluster point, so any sequence $(x_n)_{n=1}^\infty$ in A has a cluster point in A , and so A is sequentially compact. ■

This is a generalization of the Heine-Borel theorem on $X = \mathbb{R}^k$, where d is the Euclidean distance. Since $(\mathbb{R}^k, d)$ is complete, sets are closed if and only if they are complete. On the other hand, by Problem 15.4, in this space, sets are bounded if and only if they are totally bounded. Therefore, this theorem becomes:

Theorem 16.8 (Heine-Borel):

Let $A \subseteq \mathbb{R}^k$. The following are equivalent:

- (1) A is compact
- (2) A is sequentially compact
- (3) A is closed and bounded. ■

Proposition 17.1:

Let (X, d) and (Y, d') be metric spaces. Let $f : X \rightarrow Y$ be continuous, and let K be a compact subset of X . Then $f(K)$ is a compact subset of Y .

Proof:

We check that $f(K)$ is sequentially compact. So let $(y_n)_{n=1}^\infty$ be a sequence in $f(K)$. Therefore, for every $n \in \mathbb{N}$, $y_n = f(x_n)$ for some $x_n \in K$. This gives a sequence $(x_n)_{n=1}^\infty$ in K . But K is sequentially compact (since it is compact), so there exists a subsequence $(x_{n(k)})_{k=1}^\infty$ that converges (under the distance d) to a limit $x \in K$. But by continuity of f , it respects convergent sequences, so $f(x_{n(k)})$ converges to $f(x)$ as $k \rightarrow \infty$ in the distance d' . But each $x_{n(k)}$ maps to $y_{n(k)} \in f(K)$, so the subsequence $(y_{n(k)})_{k=1}^\infty$ converges to $f(x) \in f(K)$ (since $x \in K$) in the distance d' . Thus we have found a convergent subsequence of $(y_n)_{n=1}^\infty$ whose limit is in $f(K)$, so $(y_n)_{n=1}^\infty$ has a cluster point in $f(K)$, so since $(y_n)_{n=1}^\infty$ was an arbitrary sequence in $f(K)$, we see $f(K)$ is sequentially compact. ■

Definition 17.2:

A metric space (X, d) is compact if X is compact under the distance d when viewed as a subset of itself.

Theorem 17.5 (Automatic Homeomorphism):

Let (X, d) be a compact metric space, and let (Y, d') be a metric space. Let $f : X \rightarrow Y$ be a continuous and bijective function. Then (Y, d') is compact and f is a homeomorphism.

Proof:

Since f is bijective, it is surjective, so $f(X) = Y$, which is compact by Proposition 17.1 since X is compact.

Now, since f is bijective, let $g = f^{-1} : Y \rightarrow X$ be its inverse. We want to show g is continuous. Let $F \subseteq X$ be closed. We wish to verify $g^{-1}(F)$ is closed in Y . But $g^{-1}(F) = f(F)$ by bijectivity. But since F is closed and a subset of a compact set, F is compact. But since f is continuous, by Proposition 17.1, $f(F)$ is compact in Y . But compact sets are closed, so $f(F)$ is closed. Therefore $g^{-1}(F)$ is closed, so we know preimages of closed sets under g are closed, so g is continuous. ■

Theorem 17.7 (Extreme Value Theorem):

Let (X, d) be a metric space, and consider $\mathbb{R}$ under the usual distance. Let $f : X \rightarrow \mathbb{R}$ be a continuous function, and suppose $\emptyset \neq K \subseteq X$ is compact. Then there exists $a, b \in K$ such that $f(a) \leq f(x) \leq f(b)$ for all $x \in K$.

Proof:

Let $C := f(K) \subseteq \mathbb{R}$, which by Proposition 17.1 is compact since K is. Hence $C \subseteq \mathbb{R}$ is closed and bounded. Since C is bounded, we can take $\alpha := \inf(C)$ and $\beta := \sup(C)$. Since C is closed, we have $\alpha, \beta \in C$ (taking limits to the supremum and infimum and using closure). Then we have $\alpha, \beta \in C$ such that $\alpha \leq t \leq \beta$ for all $t \in C$. But $C = f(K)$, so there exists $a, b \in K$ such that $f(a) = \alpha$, $f(b) = \beta$. Thus, for any $x \in K$ and setting $t := f(x)$, the above inequality becomes $f(a) \leq f(x) \leq f(b)$. ■

Proposition 18.1:

Let X be a vector space over $\mathbb{R}$ and let $\|\cdot\|$ and $\|\cdot\|'$ be equivalent norms on X . Then $(X, \|\cdot\|)$ and $(X, \|\cdot\|')$'s Cauchy sequences coincide.

Proof:

The equivalence in the proposition is symmetric, so we only prove this once. Let $(x_n)_{n=1}^\infty$ be a Cauchy sequence in $\|\cdot\|'$, and let $\varepsilon > 0$. Apply Cauchy-ness to ε/β (where β is a Lipschitz constant on the norm equivalence) to get that $\|x_m - x_n\|' < \frac{\varepsilon}{\beta}$ for all $m, n \geq n_0$ for some $n_0 \in \mathbb{N}$. But then for any $m, n \geq n_0$, we have

$$\|x_m - x_n\| \leq \beta \|x_m - x_n\|' < \beta \frac{\varepsilon}{\beta} = \varepsilon$$

so $(x_n)_{n=1}^\infty$ is Cauchy in $(X, \|\cdot\|)$. ■

Corollary 18.2:

Let X be a vector space over $\mathbb{R}$ and let $\|\cdot\|$ and $\|\cdot\|'$ be equivalent norms on X . Then if $(X, \|\cdot\|)$ is a Banach space, so too is $(X, \|\cdot\|')$. ■

Definition 18.3/Remark 18.4:

For $k \in \mathbb{N}$, we denote

$$\mathbb{S}^{k-1} := \{x \in \mathbb{R}^k : \|x\|_2 = 1\}$$

which is called the unit sphere on $\mathbb{R}^k$. This set is compact by the Heine-Borel theorem, where it is closed since it is the boundary of the unit ball, and clearly bounded. Another property is that every vector in $x \in X$ can be uniquely written as $x = ry$ for $y \in \mathbb{S}^{k-1}$ and $r \in (0, \infty)$.

Lemma 18.5:

Let $(X, \|\cdot\|)$ be a normed vector space, and let $y_1, \dots, y_k$ be vectors in X . Then the function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ defined by

$$f(t_1, \dots, t_k) := \|t_1 y_1 + \dots + t_k y_k\|$$

for any $(t_1, \dots, t_k) \in \mathbb{R}^k$ is continuous.

Proof:

See lecture shell (TODO) ■.

Proposition 18.6:

Let X be a vector space over $\mathbb{R}$, with $\dim(X) = k < \infty$. Let $\|\cdot\|$ and $\|\cdot\|'$ be norms on X . Then $\|\cdot\|$ and $\|\cdot\|'$ are equivalent.

Proof:

Fix for the whole proof a basis $y_1, \dots, y_k$ of X . Define f as given in the above lemma (with $y_1, \dots, y_k$ our basis we use as the vectors for f), and define g to be f in the above lemma but with the norm $\|\cdot\|'$.

Then we claim there exists $\alpha_0, \beta_0 > 0$ such that for all $(t_1, \dots, t_k) \in \mathbb{S}^{k-1}$, we have $\alpha_0 \leq f(t_1, \dots, t_k) \leq \beta_0$, and similarly there exists $\alpha'_0, \beta'_0 > 0$ such that $\alpha'_0 \leq g(t_1, \dots, t_k) \leq \beta'_0$ for all $(t_1, \dots, t_k) \in \mathbb{S}^{k-1}$.

If we have these constants, then we claim we are done, since we claim this gives us

$$\frac{\alpha_0}{\beta'_0} \leq \frac{\|x\|}{\|x\|'} \leq \frac{\beta_0}{\alpha'_0}$$

for all $x \neq 0 \in X$, which we see gives us the bi-Lipschitz characterization of equivalent norms. Proving this claim, let $x \in X$ such that $x \neq 0$, and write in the basis $x = s_1 y_1 + \dots + s_k y_k$ with $s_1, \dots, s_k \in \mathbb{R}$ not all zero. Then we can write $(s_1, \dots, s_k) = r(t_1, \dots, t_k)$ for $r \in (0, \infty)$ and $(t_1, \dots, t_k) \in \mathbb{S}^{k-1}$ (normalizing it). Then we have $\|x\| = \|(rt_1)y_1 + \dots + (rt_k)y_k\| = r\|t_1 y_1 + \dots + t_k y_k\| = r f(t_1, \dots, t_k)$, giving $r\alpha_0 \leq \|x\| \leq r\beta_0$. The same calculation with $\|\cdot\|'$ and g gives $r\alpha'_0 \leq \|x\|' \leq r\beta'_0$, which dividing (the r 's cancel) gives us

$$\frac{\alpha_0}{\beta'_0} \leq \frac{\|x\|}{\|x\|'} \leq \frac{\beta_0}{\alpha'_0}$$

completing the proof of the claim.

For existence of the constants, consider the function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ given in Lemma 18.5 (where the vectors we use to define it are the basis $y_1, \dots, y_k \in \mathbb{R}^k$) defined by $f(t_1, \dots, t_k) = \|t_1 y_1 + \dots + t_k y_k\|$ for all $(t_1, \dots, t_k) \in \mathbb{R}^k$. This function is continuous, so since $S^{k-1} \subseteq \mathbb{R}^k$ is compact, by the EVT, there exist $a, b \in S^{k-1}$ such that $f(a) \leq f(t_1, \dots, t_k) \leq f(b)$ for all $(t_1, \dots, t_k) \in S^{k-1}$. Define $\alpha := f(a), \beta := f(b)$, so this gives $\alpha \leq f(t_1, \dots, t_k) \leq \beta$. Moreover, since f is just applying the norm to a vector, by non-negativity of the norm, we have $\alpha = f(a) \geq 0, \beta = f(b) \geq 0$. Moreover, since $a \in S^{k-1}$, it must have at least one nonzero component, call it a_i , where then $a_i y_i$ is not the zero vector, so $f(a) = \|a_i y_i + \text{other stuff}\| \neq 0$ because $a_i y_i + \text{other stuff}$ is not the zero vector (the other stuff cannot cancel out with $a_i y_i$ because of linear independence of the basis, which y_i is a part of). Therefore, $\alpha = f(a) > 0$, giving the final part of the result (construction of β is analogous). ■

Corollary 18.7:

Any finite dimensional normed vector space $(X, \|\cdot\|)$ is complete.

Proof:

Combine a linear isomorphism of X with $\mathbb{R}^k$ and completeness of $\mathbb{R}^k$ under your favorite ℓ^p norm. ■

Recall the space $F_b(M, \mathbb{R})$, for M a nonempty set. As previously showed, this is a Banach space under the norm $\|\cdot\|_\infty$, inducing the so-called uniform distance. Now, let us take a special case of this:

Notation 19.1/Remark 19.2:

Let (X, d) be a compact metric space. Then denote by $C(X, \mathbb{R}) := \{f : X \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$, which is a subset of $F_b(X, \mathbb{R})$ by the Extreme Value Theorem. Moreover, it is a linear subspace of $F_b(X, \mathbb{R})$, since continuous functions are preserved by scalar multiplication and pointwise addition, and so it is a normed vector space when we restrict $\|\cdot\|_\infty$ to $C(X, \mathbb{R})$. Is it a Banach space?

Proposition 19.3:

$(C(X, \mathbb{R}), \|\cdot\|_\infty)$ is a Banach space.

Proof:

We prove $C(X, \mathbb{R})$ is a closed subset of $F_b(X, \mathbb{R})$ and thus will be complete since $F_b(X, \mathbb{R})$ is (both under the distance ρ induced by $\|\cdot\|_\infty$). We verify it contains its limit points. Pick a convergent sequence $(f_n)_{n=1}^\infty$ in $C(X, \mathbb{R})$ so that $\lim_{n \rightarrow \infty} f_n = f$ (where our limit is in the distance ρ) for some $f \in F_b(X, \mathbb{R})$. We must prove f is continuous.

Fix $a \in X$ and let $\varepsilon > 0$. By convergence of $f_n \rightarrow f$ as $n \rightarrow \infty$, there exists $n_0 \in \mathbb{N}$ such that $\rho(f_n, f) < \varepsilon/3$ for all $n \geq n_0$. In particular, we have $\|f_{n_0} - f\|_\infty < \varepsilon/3$, so we get $|f_{n_0}(x) - f(x)| \leq \sup_{y \in X} |f_{n_0}(y) - f(y)| < \varepsilon/3$ for all $x \in X$. Since $f_{n_0} \in C(X, \mathbb{R})$, there is $\delta > 0$ such that for all $x \in X$ such that $d(x, a) < \delta$, we have $|f_{n_0}(x) - f_{n_0}(a)| < \varepsilon/3$. Therefore, for all $x \in X$ satisfying $d(x, a) < \delta$, we have

$$|f(x) - f(a)| \leq |f(x) - f_{n_0}(x)| + |f_{n_0}(x) - f_{n_0}(a)| + |f_{n_0}(a) - f(a)| < 3\varepsilon/3 = \varepsilon$$

where the last term is less than $\varepsilon/3$ by taking the above estimate $|f_{n_0}(x) - f(x)| \leq \sup_{y \in X} |f_{n_0}(y) - f(y)| < \varepsilon/3$ for all $x \in X$ with $x = a$. Therefore, f is continuous at arbitrary $a \in X$, so it is in $C(X, \mathbb{R})$, giving closure and thus completeness. ■

Definition 20.1:

Let (X, d) be a metric space, and let $f : X \rightarrow \mathbb{R}$ be a function. We say that f is uniformly continuous if for every $\varepsilon > 0$, we can find $\delta > 0$ such that for all $x, a \in X$ satisfying $d(x, a) < \delta$, we have $|f(x) - f(a)| < \varepsilon$.

This is a strengthening of continuity in general metric spaces, there are many easy examples of non-uniformly continuous functions on general metric spaces, but compactness gives us the converse:

Proposition 20.3:

Let (X, d) be a compact metric space, and let $f : X \rightarrow \mathbb{R}$ be continuous on X . Then f is uniformly continuous.

Proof:

Let $\varepsilon > 0$ be given. Assume by contradiction that no $\delta > 0$ works, so every $n \in \mathbb{N}$, $\delta = \frac{1}{n}$ does not work, that is, for every $n \in \mathbb{N}$, there exist $x_n, a_n \in X$ such that $d(x_n, a_n) < 1/n$ but $|f(x_n) - f(a_n)| \geq \varepsilon$. This gives a sequence $(a_n)_{n=1}^\infty$ in X , which by (sequential) compactness of X , has a convergent subsequence $(a_{n(k)})_{k=1}^\infty$ with limit $a \in X$. We claim for the same indices that $(x_{n(k)})_{k=1}^\infty \rightarrow a$. This is because for every $k \in \mathbb{N}$, we have $d(x_{n(k)}, a) \leq d(x_{n(k)}, a_{n(k)}) + d(a_{n(k)}, a) \rightarrow 0 + 0 = 0$ since the left are less than $\frac{1}{n(k)}$ by our construction of δ , and the right converges. Thus by the squeeze theorem, we have $\lim_{k \rightarrow \infty} x_{n(k)} = a$.

But now, by continuity of f , we have $\lim_{k \rightarrow \infty} f(a_{n(k)}) = a$ and $\lim_{k \rightarrow \infty} f(x_{n(k)}) = a$, so $\lim_{k \rightarrow \infty} (f(a_{n(k)}) - f(x_{n(k)})) = a - a = 0$, contradicting $|f(a_{n(k)}) - f(x_{n(k)})| \geq \varepsilon$ for all $k \in \mathbb{N}$. ■

Definition 20.4:

Let (X, d) be a compact metric space, and let $\mathcal{B} \subseteq C(X, \mathbb{R})$. We say $\mathcal{B}$ is equicontinuous if for every $\varepsilon > 0$, there is $\delta > 0$ such that if $x, a \in X$ satisfy $d(x, a) < \delta$, we have $|f(x) - f(a)| < \varepsilon$ for all $f \in \mathcal{B}$.

Definition 20.6:

Let (X, d) be a compact metric space, and let $\mathcal{B} \subseteq C(X, \mathbb{R})$. We say $\mathcal{B}$ is uniformly bounded if $|f(x)| \leq r$ for all $x \in X$ and $f \in \mathcal{B}$, or equivalently, $\|f\|_\infty \leq r$ for all $f \in \mathcal{B}$.

Theorem 20.7 (Arzela-Ascoli):

Let (X, d) be a compact metric space and let $\mathcal{B} \subseteq C(X, \mathbb{R})$. Then $\mathcal{B}$ is totally bounded in $(C(X, \mathbb{R}), \rho)$ if and only if $\mathcal{B}$ is equicontinuous and uniformly bounded.

Proof:

In the forward direction, let $\varepsilon > 0$. By total boundedness, there is a finite $\varepsilon/3$ -net $f_1, \dots, f_p \in \mathcal{B}$ for $\mathcal{B}$, that is, for every $f \in \mathcal{B}$, there exists $1 \leq i \leq p$ such that $\|f - f_i\|_\infty < \varepsilon/3$. Furthermore, by Proposition 20.3, this f_i is uniformly continuous because $f_i \in \mathcal{B} \subseteq C(X, \mathbb{R})$, where $C(X, \mathbb{R})$ is compact. Therefore, for every $1 \leq i \leq p$, there exists $\delta_i > 0$ such that for all $x, a \in X$ satisfying $d(x, a) < \delta_i$, we have $|f_i(x) - f_i(a)| < \varepsilon/3$.

Now, let $\delta := \min(\delta_1, \dots, \delta_p) \in (0, \infty)$. We claim this works. Pick a function $f \in \mathcal{B}$ and let $x, a \in X$ satisfy $d(x, a) < \delta$. Then,

$$\begin{aligned} |f(x) - f(a)| &\leq |f(x) - f_i(x)| + |f_i(x) - f_i(a)| + |f_i(a) - f(a)| \\ &\leq \|f - f_i\|_\infty + |f_i(x) - f_i(a)| + \|f - f_i\|_\infty < \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon \end{aligned}$$

Therefore, we see $\mathcal{B}$ is equicontinuous. Uniform boundedness is clear from total boundedness, since uniform boundedness just means $\mathcal{B}$ is bounded in $(C(X, \mathbb{R}), \rho)$ but $\mathcal{B}$ is totally bounded, which is stronger. This completes the proof of the forward direction. The backwards direction can be found in an appendix (TYPE LATER). ■

Corollary 20.8:

Let (X, d) be a compact metric space and let $\mathcal{B} \subseteq C(X, \mathbb{R})$. Then $\mathcal{B}$ is compact in $(C(X, \mathbb{R}), \rho)$ if and only if $\mathcal{B}$ is equicontinuous, uniformly bounded, and closed.

Proof:

In the forward direction, since $\mathcal{B}$ is compact, it is complete and totally bounded, where the above theorem shows totally bounded gives equicontinuity and uniform boundedness, and completeness gives closure.

In the reverse direction, since $C(X, \mathbb{R})$ is complete, a closed subset of it is complete, and then the rest of the result follows from the above theorem. ■

A typical application:

Problem P20.1:

Let (X, d) be $[0, 1]$ under the usual distance. Suppose $f_1, \dots, f_n, \dots$ be a sequence of functions in

$C([0, 1], \mathbb{R})$ such that $f_n(0) = 0$ for all $n \in \mathbb{N}$ and $|f_n(s) - f_n(t)| \leq \sqrt{t-s}$ for all $n \in \mathbb{N}$ and $0 \leq s \leq t \leq 1$. Then we can find a subsequence $f_{n(1)}, \dots, f_{n(k)}, \dots$ such that $f_{n(k)}$ uniformly converges to a function $f \in C([0, 1], \mathbb{R})$.

Solution:

Let $\mathcal{B} = \{f \in C([0, 1], \mathbb{R}) \mid |f(s) - f(t)| \leq \sqrt{t-s} \text{ for all } 0 \leq s \leq t \leq 1 \text{ and } f(0) = 0\}$. Use corollary 20.8 to get compactness (these verifications are supposed to be pretty easy), getting sequential compactness and completing the proof. ■

There is a classical theorem of Weierstrass:

Theorem 21.1 (Weierstrass Approximation Theorem):

Let $[a, b] \subseteq \mathbb{R}$ be a nonempty interval, and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then, for every $\varepsilon > 0$, there exists a polynomial p such that $|f(t) - p(t)| < \varepsilon$ for all $t \in [a, b]$.

We can restate this as:

Theorem 21.2:

Consider the metric space $(C([a, b], \mathbb{R}), \rho)$. Let $\mathcal{P} \subseteq C([a, b], \mathbb{R})$ be the set of polynomial functions whose domain is restricted to $[a, b]$. Then $\mathcal{P}$ is dense in $(C([a, b], \mathbb{R}), \rho)$.

To see this, we have (for all $t \in [a, b]$) $|f(t) - p(t)| < \varepsilon$ if and only if $\|f - p\|_\infty = \sup_{t \in [a, b]} \{|f(t) - p(t)|\} = \max_{t \in [a, b]} \{|f(t) - p(t)|\} < \varepsilon$ if and only if $p \in B(f; \varepsilon)$ in the metric space $(C([a, b], \mathbb{R}), \rho)$. Thus in the notation of the above theorem, we see $\mathcal{P} \cap B(f; \varepsilon) \neq \emptyset$ for all $f \in C([a, b], \mathbb{R})$ and all $\varepsilon > 0$.

Remark 21.4:

In addition to being a vector space, $C(X, \mathbb{R})$ also has multiplication defined pointwise, satisfying some obvious axioms when combined with its vector addition and scalar multiplication, with an identity $\underline{1} : X \rightarrow \mathbb{R}$ given by $\underline{1}(x) = 1$ for all $x \in X$. Such a structure is called a (unital) algebra, and we say that $C(X, \mathbb{R})$ is a (unital) algebra of functions.

Definition 21.5:

Let $\mathcal{A}$ be a subset of $C(X, \mathbb{R})$. We say that $\mathcal{A}$ is a unital subalgebra of $C(X, \mathbb{R})$ if it satisfies the following conditions:

- (i) $\underline{1} \in \mathcal{A}$
- (ii) $[f, g \in \mathcal{A}] \implies [f + g \in \mathcal{A}]$
- (iii) $[f \in \mathcal{A}, \alpha \in \mathbb{R}] \implies [\alpha \cdot f \in \mathcal{A}]$
- (iv) $[f, g \in \mathcal{A}] \implies [fg \in \mathcal{A}]$.

Definition 21.6:

Let $\mathcal{S}$ be a subset of $C(X, \mathbb{R})$. If for every $x_1 \neq x_2$ in X there exists a function $f \in \mathcal{S}$ such that $f(x_1) \neq f(x_2)$, then we say $\mathcal{S}$ separates the points of X .

Theorem 21.7 (Stone-Weierstrass):

Let (X, d) be a compact metric space, and let $\mathcal{A} \subseteq C(X, \mathbb{R})$ be a unital subalgebra of $C(X, \mathbb{R})$ that separates the points in X . Then $\mathcal{A}$ is dense in $(C(X, \mathbb{R}), \rho)$ where ρ is the uniform distance associated to $\|\cdot\|_\infty$.

Remark 21.8:

The above theorem indeed generalizes Weierstrass's Approximation Theorem (21.1): if we take $X = [a, b] \subseteq \mathbb{R}$, d to be the usual distance, and $\mathcal{A}$ to be the set of polynomial functions on $[a, b]$, then $\mathcal{A}$ is indeed a unital subalgebra of $C([a, b], \mathbb{R})$ that separates points, and so is dense in $C([a, b], \mathbb{R})$, that is, we can approximate continuous functions on $[a, b]$ arbitrarily well with polynomials, which is exactly what Theorem 21.1 says!

Problem P21.3:

A function $u : [0, 2\pi] \rightarrow \mathbb{R}$ is said to be a trigonometric polynomial when it can be written in the form

$$u(x) = \sum_{k=0}^n \alpha_k \cos kx + \sum_{k=1}^n \beta_k \sin kx \quad \forall x \in [0, 2\pi]$$

for some $n \geq 0$ and coefficients $\alpha_0, \alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n \in \mathbb{R}$. Then every continuous function $f : [0, 2\pi] \rightarrow \mathbb{R}$ can be approximated uniformly with trigonometric polynomials. That is, given a continuous function f satisfying $f(0) = f(2\pi)$, and $\varepsilon > 0$, there exists a trigonometric polynomial u such that $|f(x) - u(x)| < \varepsilon$ for all $x \in [0, 2\pi]$.

Proof:

We want to apply Stone-Weierstrass. It's not too hard to check with trigonometry that the set of trigonometric polynomials is a unital subalgebra of $C([0, 2\pi], \mathbb{R})$ (the main difficulty is checking it is closed under vector multiplication). The main problem is that they don't separate all points, by periodicity, they don't separate 0 from 2π ! So let's choose a different metric space X . The idea is to use complex numbers. Let $X = \mathbb{T}$, the unit circle in the complex plane (under the usual distance $d(z, w) = |z - w|$). This is a compact subset of $\mathbb{C}$, which can be seen by identifying $\mathbb{C}$ and $\mathbb{R}^2$, under which $\mathbb{T}$ is the previously-seen-compact set $\mathcal{S}^1$.

Let $\mathcal{C} = \{v : \mathbb{T} \rightarrow \mathbb{R} \mid \exists \alpha_0, \alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n \in \mathbb{R} \text{ s.t. } v(z) = \alpha_0 + \sum_{k=1}^n \alpha_k \operatorname{Re}(z^k) + \sum_{k=1}^n \beta_k \operatorname{Im}(z^k)\}$. We can check this is a unital subalgebra of $C(\mathbb{T}, \mathbb{R})$ that separates points. The latter holds because $v_1(z) = \operatorname{Re}(z)$ and $v_2(z) = \operatorname{Im}(z)$ are in this set, which collectively separate the points in $\mathbb{T}$. Therefore, $\mathcal{C}$ is dense in $C(\mathbb{T}, \mathbb{R})$.

Now suppose we are given $f : [0, 2\pi] \rightarrow \mathbb{R}$ continuous with $f(0) = f(2\pi)$ and $\varepsilon > 0$. Let $g : \mathcal{T} \rightarrow \mathbb{R}$ be defined via the condition that $g(\cos x + i \sin x) = f(x)$ for all $0 \leq x \leq 2\pi$. This is well-defined and continuous due to the assumption that $f(0) = f(2\pi)$.

Use Stone-Weierstrass to conclude that there exists $v \in \mathcal{C}$ such that $|g(z) - v(z)| < \varepsilon$ for all $z \in \mathcal{T}$. Let $u : [0, 2\pi] \rightarrow \mathbb{R}$ be defined by $u(x) = v(\cos x + i \sin x)$ for all $0 \leq x \leq 2\pi$. This is indeed a trigonometric polynomial (by De Moivre's Formula), and by now substituting u and f into our above estimate, we get $|u(x) - f(x)| < \varepsilon$ for all $x \in [0, 2\pi]$. ■

Remark:

The proof of Theorem 21.7 due to Stone was based upon subsets $\mathcal{L} \subseteq C(X, \mathbb{R})$ called lattices of functions, which are structures that satisfy $f, g \in \mathcal{L}$ implies $f \vee g \in \mathcal{L}$ and $f, g \in \mathcal{L}$ implies $f \wedge g \in \mathcal{L}$, where $f \vee g = \max_{x \in X} (f(x), g(x))$ and $f \wedge g = \min_{x \in X} (f(x), g(x))$.